The 043 Datasets

Fifteen expression, protein and morphology datasets: genotypes, agreement, co-location, and the design of one model over every gene-level readout

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Abstract

The 043 store holds 173,617 records of 15 yeast datasets that read mRNA, protein abundance or turnover, or CalMorph morphology, kept as served with no aggregation. Its genotypes fall into four classes, and today a model sees only one of them as intended: deletions are gene sets, isolates mark almost every gene as perturbed, segregants mark none, and the promoter library has no gene-level genotype. Each dataset is set against its own reproducibility. Kemmeren 2014's two cultures of 114 mutants give an own-culture ceiling of 0.729; it replaces the experiment 019 ceiling, which rested on Sameith 2015's single mutants, found here to be Kemmeren's own arrays. Two constructions of one Jackson 2020 genotype agree on the deletion effect at a median r of 0.129 against 0.095 for different genotypes; Hu 2007 released no replicate; no segregant appears in two panels. Between datasets, agreement is low (median per-genotype r from -0.006 to 0.208 apart from the Sameith copy), and morphology is the label most often shared with expression and protein. The plan trains every gene-level readout in one model and asks two engineering questions of it: held-out Kemmeren single deletions by gene, and all 72 Sameith double deletions against the additive null of their singles; the Muenzner proteome is a third test and Ohya morphology a probe of the frozen latent. The data as tensors, the input side (gene tokens, change-type marks and the perturbation operator) and the output side (readout sites, heads, tokens, losses and the evaluation protocol) follow, with the choices left to the owner collected in one table.

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1 The 043 store ✗

Query. The store is one query over the served knowledge graph, KG 4.0 (release `2026.10.17-4295f7ca`), registered as the supported query `expression_proteome_morphology_043`. It takes every served yeast dataset whose readout is a vector of mRNA expression, protein abundance or turnover, or CalMorph morphology: 15 datasets, 173,805 served experiments. Five candidates were left out: Puddu 2019 holds a realized-genotype table rather than a cell-state readout, Ohnuki 2018 and 2022 share no strain with the other sets, Hackett 2020 carries an induction the graph processor would read as a deletion, and Messner 2021 carries no genotype perturbation.

The keep-the-data rule. The store holds one sample per served pair of experiment and experiment reference, with no unit converter, no deduplicator and no aggregation over genotypes. Two reasons were measured on the code: the mean deduplicator keys on the perturbed gene names alone, so it would average one deletion across Jackson 2020's eleven conditions, and it raises on four of the store's experiment types. Reconciling units and replicates is therefore a modeling decision, taken per batch, rather than a property of the store.

Size. The raw store holds 173,617 samples. The 188 served experiments it lacks name a gene outside the genome's 6,607-gene set and fail the deletion blocks' clause that every perturbed gene be in that set: Kemmeren 2, Hu 1, Nadal-Ribelles 70, Messner 75, Ohya 40 (slurm job 5855; `results/store_census.json`).

Terms. Table 1 fixes the words the rest of the text uses. Table 2 gives each dataset with the statistic that bounds what a model can reach on it.

Table 1: Controlled vocabulary, alphabetical.

Term	Meaning
Ceiling	The per-gene Pearson a perfect predictor of the true signal would score against one measurement: $\sqrt{\text{rel}_g}$, averaged over genes, with rel_g the share of variance across strains that is signal. It is a bound on a model, not a score of one.
Construction	One independently built strain of a genotype. Jackson 2020 built six per genotype, each with its own transcribed barcode, and the store keeps each as its own record.
Co-location	Two labels measured on the same genotype key, in any datasets.
Deletion-effect profile	Per gene, $\log_2((\text{CPM}+1)/(\text{CPM}_{\text{HO}}+1))$ for a construction against the HO-deletion control of the same condition.
Haplotype block	A run of markers on one chromosome over which a segregant's called parent and that call's posterior are constant; the genotype of a segregant is a list of blocks against two pinned parent assemblies.
Isolate	A natural strain; its genotype is thousands of sequence variants, gene presences and absences, and natural loss-of-function calls against the reference.
Pseudobulk	The UMI counts of several single cells summed per gene.
Segregant	A haploid meiotic product of a cross between two parent strains. Pooled segregants were genotyped by sequencing before profiling; de novo segregants are genotyped from each cell's own transcripts.
Test-retest	Two independent measurements of the same strains, correlated across strains per gene (r_g, so $\text{rel}_g = r_g$) or across genes per strain.

Table 2: The 15 datasets of the 043 store. Genotypes and records count the raw store; features per record is the median length of the stored label vector (genes, proteins or CalMorph features; one scalar for Vaishnav). Environments count distinct media, temperature and environmental perturbation; temperature is a typed gap where the paper states none. The last column gives each dataset's agreement with itself and the statistic used: a ceiling is the mean over genes of $\sqrt{\text{rel}_g}$ (Sec. 3.1); Jackson is the median Pearson between two constructions of one genotype against two of different genotypes, Nadal-Ribelles the median Pearson of one genotype across batches against different genotypes. Messner, Zelezniak, Ohya, Caudal and Nadal-Ribelles were measured in experiments 019 and 028, before this store.

Dataset	Genotype class	Readout, scale	Genotypes, n	Records, n	Features per record, median	Environments, n: name	Within-dataset agreement, statistic in cell
Kemmeren 2014	deletion	two-color array, log2 ratio to WT pool	1,482	1,482	6,169	1: SC, 30 C	0.73 own-culture ceiling, per gene (114 mutants with 2 cultures)
Sameith 2015 Sm	deletion	two-color array, log2 ratio to WT pool	82	82	6,169	1: SC, 30 C	none: the same arrays as Kemmeren (82 of 82 identical)
Sameith 2015 Dm	double deletion	two-color array, log2 ratio to WT pool	72	72	6,169	1: SC, 30 C	not measured
Hu 2007	deletion	error-model X score vs WT, signed	262	262	6,381	1: YPD, temperature a typed gap	not measurable: replicates averaged before release; Kemmeren × Hu r 0.11 is the bound
Jackson 2020	deletion, 6 construc-tions	scRNA-seq UMI counts, summed per construction	12	784	4,028	10: 9 media	construction pair r 0.13 vs 0.09 null, effect, floor 100
Nadal-Ribelles 2025	deletion	pseudobulk log2 fold change vs WT	3,059	6,118	5,866	2: 1 media	same genotype across batches r 0.043 vs 0.022 null
Messner 2023	deletion	SWATH protein abundance, linear	4,476	4,624	1,819	1: SM, 30 C	0.70 ceiling, 388 WT replicates
Zelezniak 2018	deletion	SWATH protein abundance, log	97	97	726	1: SM, 30 C	0.92 ceiling, per-strain replicate SE
Ohya 2005	deletion	CalMorph features, per feature	4,678	4,678	281	1: YPD, 25 C	0.61 ceiling, 122 WT replicates
Caudal 2024	isolate	RNA-seq TPM, linear	943	943	6,050	1: SC, 30 C	0.60 ceiling, 29 re-cultured isolates
Muenzner 2024 protein	isolate	SWATH protein abundance, linear	795	795	1,540	1: SM, 30 C	not measured
Muenzner 2024 turnover	isolate	SILAC degradation rate, per hour	55	55	1,055	1: SM + 80 mg/L [13C6/15N2] L-lysine, 30 C	not measured
Boocock 2025	segregant	scRNA-seq UMI counts, per cell or per segregant	79,516	79,516	962	2: SM, 30 C; 2x YNB + 2% glucose added to FACS sheath fluid, 30 C	no segregant in two panels; nearest-genotype deviation r 0.11 vs 0.07 random
N'Guessan 2025	segregant	scRNA-seq UMI counts, summed per segregant	2,951	2,951	2,592	1: YPD, temperature a typed gap	not measurable: cells summed per segregant, no replicate
Vaishnav 2022	promoter library	sort-seq expression level, one scalar	1	71,158	1	1: SC-Ura, 30 C	not measured
Total				173,617			

2 Genotype classes and how each enters a model ✗

The store holds four genotype classes, and the graph processors that turn a record into model input treat them differently. What follows is the behavior of the code at the 043 build (`torchcell/data/graph_processor.py`), measured on the processed samples.

Deletions. Kemmeren, Sameith, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak and Ohya: every record names one or two deleted genes (two for Sameith Dm). The genotype is the set of systematic names, and the `Perturbation` processor marks those nodes in `pert_mask`. Jackson 2020 adds a construction identity (plate well and barcode) below the gene name, so its 784 records hold 72 constructions of 12 genotypes.

Isolates. Caudal 2024 and the two Muenzner 2024 datasets: a natural isolate's genotype is its variant calls against S288C, five perturbation types with the counts of Table 3, between 195 and 7,025 perturbations per Caudal record. The `Perturbation` processor marks every gene with any of them, so a Caudal sample has 5,575 of 6,607 nodes in `pert_mask`: the mask then separates almost nothing, and a rule over perturbation types (for example loss of function and absence only) is needed before isolates are trained on.

Table 3: Perturbations of an isolate record, by type. Mean count per record over the records of each dataset (`results/store_census.json`). Sequence variants are named by the gene they fall in; natural loss of function, gene absence and gene presence are released calls; the few deletions are engineered markers.

Dataset	Records, n	Perturbations per record, min to max	Sequence variant, mean per record	Natural LoF, mean per record	Gene absence, mean per record	Gene presence, mean per record	Deletion, mean per record
Caudal 2024	943	195 to 7,025	5,047	383	135	84	0
Muenzner 2024 protein	795	0 to 7,025	5,017	377	133	84	0
Muenzner 2024 turnover	55	4,903 to 6,430	5,059	373	134	90	0

Segregants. Boocock 2025 and N'Guessan 2025: a `SegregantGenotype` names a cross, two parent strains pinned to sha256-verified assemblies, and a list of haplotype blocks, and it carries no perturbation list by design. Since the stage-2 fix the processors read a missing list as zero perturbed genes, so a segregant sample today has `pert_mask` all false: a model sees it as the reference strain. Using segregants needs a representation of the blocks; Sec. 5 names one.

Promoter library. Vaishnav 2022: each of 71,158 records is one 80 bp variable promoter insert driving a YFP reporter, read as one scalar. Every record names the same reporter, so the store holds one gene-name genotype for all of them; the perturbation sets separate them only by sequence content. No gene-level vector and no gene-set encoder applies.

3 Agreement within and between datasets ✗

3.1 Within a dataset ✗

A dataset's agreement with itself bounds both what a model can score on it and what any other dataset can agree with it. The new measurements are in `experiments/043-expression-proteome-morphology/scripts/within_dataset_ceilings.py` (slurm jobs 5893 for Kemmeren and Hu, 5895, 5896 and 5898 for Jackson, the segregants and the figures).

Jackson 2020. Each of the 784 records is one construction in one condition, a median of 23 cells and 43,851 UMIs (IQR 19,757 to 101,152); YPD and rapamycin reach a median of about 400,000 UMIs per construction, the other nine conditions 14,000 to 93,000. Within one condition, two constructions of the same genotype agree on their deletion-effect profiles only slightly better than two constructions of different genotypes (Table 4, Fig. 1a): a median r of 0.129 against 0.095 with every gene of at least 100 UMIs in the condition (1,778 same-genotype and 21,300 different-genotype pairs). The gap widens on the 660 deepest genes (0.244 against 0.103 at 1,000 UMIs) and when only genes covered in both constructions are kept, but that second rule scores fewer pairs, the deep ones. The different-genotype pairs are not near zero because both constructions are divided by the same HO reference. The Jackson ceiling for one construction's deletion effect is therefore low; most of a record's deviation from HO is count noise shared with any other construction in the flask.

Kemmeren 2014. GEO titles name the culture: `bck1-del-1-a` and `-1-b` are the dye-swapped arrays of culture 1. Of 1,480 mutants whose titles parse, 114 carry two cultures, 72 in the responsive series GSE42527 and 42 in the non-responsive series GSE42526; the other 1,366 have one (16 arrays with irregular titles are left out). Culture 1 against culture 2 of the same mutant gives a median per-strain r of 0.730, against 0.284 for culture 1 of one mutant and culture

Table 4: Jackson 2020, construction against construction within one condition. Pooled over the 11 conditions. Level profile: Pearson over genes of $\log_2(\mathrm{CPM}+1)$; deletion effect: of $\log_2((\mathrm{CPM}+1)/(\mathrm{CPM}_{\mathrm{HO}}+1))$ against the six HO constructions summed, HO pairs left out. A gene enters every pair of a condition when its UMIs summed over that condition reach the floor. Same genotype: two of the six constructions of one genotype; different: two constructions of different genotypes in the same flask.

Gene rule	Level profile r, median, same vs different genotype	Deletion-effect r, median, same vs different genotype	Pairs, n, same / different	Genes per pair, median
$\geq$ 10 UMIs summed over the condition	0.637 vs 0.620	0.184 vs 0.157	1,778 / 21,300	5,405
$\geq$ 100 UMIs summed over the condition	0.654 vs 0.612	0.129 vs 0.095	1,778 / 21,300	3,701
$\geq$ 1000 UMIs summed over the condition	0.919 vs 0.887	0.244 vs 0.103	1,778 / 21,300	660
$\geq$ 10 UMIs in both constructions	0.941 vs 0.925	0.406 vs 0.244	1,742 / 20,866	490
$\geq$ 100 UMIs in both constructions	0.968 vs 0.951	0.414 vs 0.164	1,191 / 12,337	34
$\geq$ 1000 UMIs in both constructions	0.997 vs 0.991	0.600 vs 0.112	271 / 2,924	0

2 of another (Table 5, Fig. 1b). Across the 114 mutants the median per-gene r is 0.612, so the own-culture ceiling is 0.729 by test-retest and 0.653 by decomposition (Fig. 1c). The two differ because the 114 paired mutants are enriched for strong responders, while the decomposition takes its signal variance from all 1,480. These numbers replace the void 0.775. The stored record averages both cultures where two exist; hypothesis, untested, that its ceiling is higher than the single-culture one.

Table 5: Kemmeren 2014, culture 1 against culture 2. A culture's profile is the mean over its dye-swapped arrays of the within-array $\log_2$(deletion/reference pool), computed by the loader's own functions from the GEO arrays of GSE42527 and GSE42526. The null pairs culture 1 of one mutant with culture 2 of another. Per-gene ceilings average over genes with at least three paired mutants; the decomposition takes the noise variance from the paired cultures and the total variance from the first culture of every mutant.

Kemmeren 2014 own-culture statistic	Value
Mutants with two cultures, n (responsive / non-responsive series)	114 of 1,480 (72 / 42)
Per-strain r across genes, median [IQR]	0.730 [0.602, 0.835]
Per-strain r, different mutants (null), median	0.284
Per-gene r across mutants, median [IQR]	0.612 [0.431, 0.754] (6,169 genes)
Ceiling, test-retest: mean over genes of $\sqrt{r_g}$	0.729
Ceiling, decomposition: mean over genes of $\sqrt{1-\sigma^2_{e,g}/\sigma^2_g}$	0.653
Void 019 cross-study ceiling (Sameith pair, identical arrays)	0.775

Hu 2007. No replicate statistic is computable. The paper says "Each experiment was independently carried out at least twice" (paper.md line 55), and the replicates were combined before release: the group "performed weighted duplicate averaging with custom Java software" (line 59). GEO GSE4654 holds one X score per strain; the per-array data went to the Longhorn Array Database and a lab site that was never archived. The only available bound is the agreement with Kemmeren, a median per-genotype r of 0.107 over 255 deletions and a per-gene r of 0.097 (Table 7). That agreement is a floor on Hu's reproducible signal; no ceiling is measured.

Boocock 2025 and N'Guessan 2025. Neither panel has a replicate: every pooled record sums the cells of one segregant (a median of 17 for Boocock Ap, 3 for N'Guessan), and the per-cell counts are not stored, so no split-half is possible (Table 6). No segregant appears in two panels: on a 5 kb grid of called parents the best concordance of any of the 393 pooled segregants with any of 27,744 de novo BY × RM cells is a median of 0.672, and with any of the 2,951 N'Guessan segregants 0.657, where unrelated segregants share half the genome and identity would be near 1; none reaches 0.95 (Fig. 2b). A weaker test still asks whether genotype carries expression across panels. For each pooled segregant, the 20 de novo cells nearest in genotype (mean concordance 0.647) agree with its deviation from the panel mean at a median r of 0.113, against 0.070 for 20 random cells; the paired difference is 0.040 (388 segregants, 63% positive, Wilcoxon $p = 1 \times 10^{-8}$). The single nearest N'Guessan segregant gives 0.101 against 0.060 (381 segregants, $p = 0.009$). At the panel level the summed BY × RM profiles agree at $r = 0.872$ between Boocock's pooled and de novo panels and 0.693 between Boocock's pooled panel and N'Guessan, which grew in YPD rather than minimal medium. Genotype posteriors differ by panel: a median 0% of markers of a pooled Boocock segregant and 1.7% of an N'Guessan one are called with posterior below 0.9, against 24 to 35% for a de novo cell.

3.2 Between datasets ✗

Records are matched by perturbed gene names and environment (isolates by strain name), and each pair is scored per shared genotype, across shared genes, and per shared gene, across shared genotypes (Table 7, Fig. 3; `experiments/043-expression-proteome-morphology/scripts/cross_dataset_correlation.py`, slurm jobs 5890 and 5892). Every pair compares log against log, so the base of a logarithm does not change a correlation.

Table 6: The segregant panels. A pooled record sums the cells matched to one previously genotyped segregant; a de novo record is one cell genotyped from its own transcripts. The posterior column is the marker-weighted fraction of a genotype whose called parent has posterior below 0.9.

Panel	Records, n	Cells, n	Cells per record, median	UMIs per record, median	Genes detected per record, median	Marker fraction with posterior < 0.9, median
Boocock Ap, pooled BY × RM	393	7,124	17	27,902	3,595	0.000
Boocock A, de novo BY × RM	27,744	27,744	1	4,017	1,228	0.236
Boocock B, de novo YJM145 × YPS163	44,784	44,784	1	2,130	859	0.352
Boocock 3004, de novo CBS2888 × YJM981	6,595	6,595	1	1,982	605	0.297
N'Guessan, pooled BY × RM	2,951	14,263	3	18,221	2,592	0.017

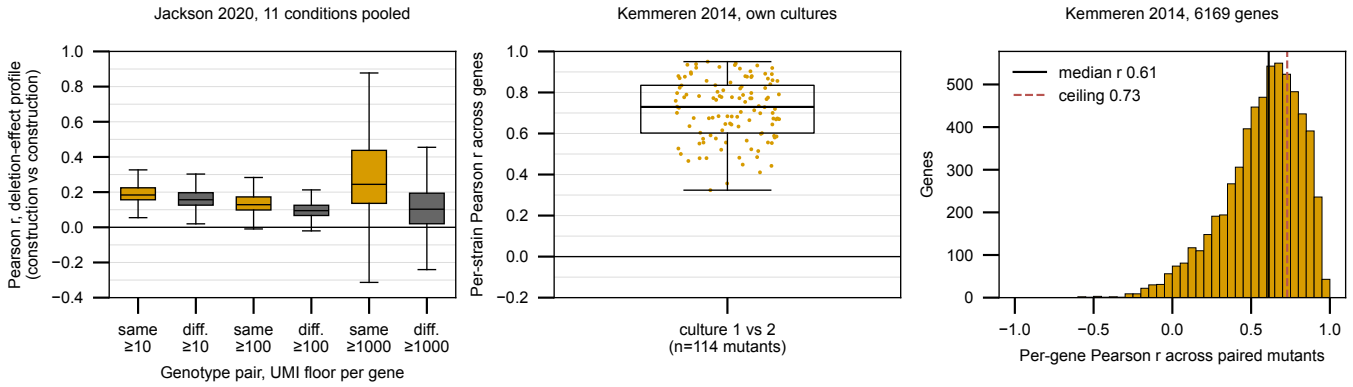


Figure 1: Two datasets measured against themselves. **a**, Jackson 2020: Pearson over genes of the deletion-effect profiles of two constructions in one condition, same genotype (orange) against different genotypes (gray), at three floors on a gene's UMIs summed over the condition; 1,778 and 21,300 pairs per floor, 11 conditions pooled, HO pairs left out. **b**, Kemmeren 2014: Pearson across genes between the two cultures of each of the 114 mutants profiled twice; culture 1 of one mutant against culture 2 of another gives a median of 0.284 (not drawn). **c**, per-gene Pearson across the 114 mutants for 6,169 genes, with the median r and the ceiling, the mean over genes of $\sqrt{\max(r_g, 0)}$.

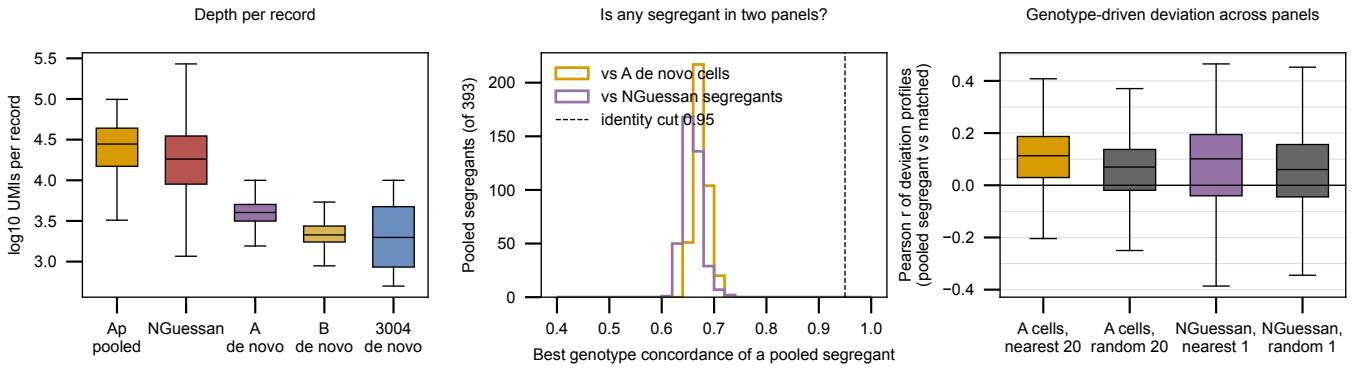


Figure 2: The segregant panels have no shared segregant and a weak shared genotype signal. **a**, UMIs per record by panel (pooled records sum a segregant's cells, de novo records are single cells). **b**, for each of the 393 pooled Boocock segregants, the best genotype concordance with any de novo cross-A cell (orange) or any N'Guessan segregant (purple), on a 5 kb grid of called parents; the dashed line marks 0.95. **c**, Pearson over genes between a pooled segregant's deviation from its panel and that of its 20 nearest de novo cells, 20 random cells, its nearest N'Guessan segregant, or a random one; genes with at least 10 UMIs on both sides.

Table 7: Agreement between datasets of the same readout type. Per genotype: Pearson and Spearman across the shared genes (at least 20). Per gene: Pearson across the shared genotypes (at least 5). Jackson is turned into a deletion effect against HO in the same environment; Messner and Zelezniak are log ratios to their record reference; Caudal and Muenzner are centered per gene over the isolates. The Jackson environment row pairs one genotype across two environments (605 genotype-environment pairs).

Pair	Shared genotypes, n	Per-genotype Pearson r, median [IQR]	Per-genotype Spearman, median	Per-gene Pearson r, median (genes, n)	Genes with per-gene r > 0.3, fraction
Kemmeren × Sameith Sm	82	1.000 [1.000, 1.000]	1.000	1.000 (6,169)	1.000
Kemmeren × Hu	255	0.107 [0.013, 0.184]	0.082	0.097 (5,803)	0.102
Kemmeren × Nadal, YPD	913	0.002 [−0.017, 0.022]	0.007	0.017 (5,891)	0.007
Kemmeren × Nadal, NaCl	920	−0.001 [−0.021, 0.022]	0.005	0.015 (5,896)	0.006
Kemmeren × Jackson, YPD 10 h	11	0.043 [0.026, 0.070]	0.047	0.035 (5,340)	0.247
Kemmeren × Jackson, YPD 4 h (a)	11	0.066 [0.027, 0.210]	0.030	0.118 (5,353)	0.321
Kemmeren × Jackson, YPD 4 h (b)	11	0.076 [0.036, 0.149]	0.045	0.052 (5,601)	0.253
Nadal YPD × Nadal NaCl	2,952	0.208 [0.168, 0.231]	0.043	0.183 (6,134)	0.050
Nadal duplicates, one condition	105	0.376 [0.323, 0.452]	0.178	0.062 (5,897)	0.089
Jackson, environment × environment	605	0.010 [−0.005, 0.029]	0.009	n/a	n/a
Messner × Zelezniak (protein)	89	0.049 [−0.016, 0.185]	0.050	0.085 (715)	0.067
Kemmeren mRNA × Messner protein	1,349	0.031 [−0.025, 0.112]	0.034	0.094 (1,832)	0.033
Caudal mRNA × Muenzner protein	759	0.208 [0.109, 0.307]	0.176	0.160 (1,572)	0.183
Muenzner turnover × abundance	55	−0.006 [−0.053, 0.047]	−0.003	−0.007 (1,120)	0.066

Sameith 2015 single mutants are Kemmeren 2014's arrays. All 82 Sameith Sm profiles are value-for-value identical to Kemmeren's for the same deletions (median per-genotype $r = 1.000$), and the GEO deposits show why. GSE42536 holds 144 Sameith single-mutant arrays: 115 are byte-identical re-deposits of GSE42527 or GSE42526 arrays, and the other 29 match a Kemmeren array exactly on the normalized Cy5 and Cy3 signal columns. Sameith's Methods defer to the earlier paper: "Full details of all gene expression profiling procedures have been described before [46]". The pair is therefore no replicate, and the experiment 019 expression ceiling built on it (test-retest 0.744 per strain, ceiling 0.77) is void; Table 5 replaces it. The 143 Sameith double-mutant arrays are new hybridizations.

Nadal-Ribelles 2025. Nadal-Ribelles agrees with Kemmeren at a median per-genotype r of 0.002 in YPD (913 deletions) and with itself not much better. Experiment 028 measured the same genotype across batches at a median r of 0.043 against 0.022 for different genotypes, and split halves at 0.080 against 0.051. The 0.376 of within-condition duplicates in Table 7 is Pearson driven by a few extreme log fold changes (the Spearman of the same pairs is 0.178); hypothesis, untested, that they come from genes with near-zero counts on one side.

Jackson 2020 across environments. One TF deletion's effect profile in one environment agrees with the same deletion's profile in another at a median r of 0.010 (605 pairs). The paper reports that TF deletions change far fewer genes than conditions do (96 differentially expressed genes in YPD, 160 in rapamycin), so a near-zero cross-environment agreement is what the paper leads one to expect; it does not show the data to be noise, and the within-condition construction pairs above measure the noise directly.

mRNA against protein. Deletions agree weakly: Kemmeren against Messner gives a median per-genotype r of 0.031 over 1,349 deletions and a per-gene r of 0.094 over 1,832 genes. Isolates agree more: Caudal against Muenzner gives 0.208 over 759 isolates and 0.160 per gene. Protein varies far more across isolates than across deletions (Fig. 4, left, x axes); hypothesis, untested, that this larger range is why the two readouts agree more there. Turnover against abundance is near zero (55 isolates).

Expression to morphology. A ridge from each gene-level matrix to the 281 CalMorph features of the shared Ohya genotypes (5-fold out-of-fold, one row-permutation null) is the one cross-label link that matters for a held-out probe (Table 8, Fig. 4, right). Kemmeren carries the most signal per genotype; Messner reaches nearly as far on three times the genotypes.

Table 8: Ridge from a gene-level matrix to CalMorph. Out-of-fold Pearson per CalMorph feature across held-out genotypes; inputs and targets z-scored and clipped at 8, penalty chosen on an inner split. The 116 moving features are the experiment 019 list (`calmorph_moving_features.json` on the multimodal-phenotype-retrospective worktree).

Input matrix	Genotypes shared with Ohya, n	Mean r over 281 features	Row-permutation null, mean r	Median r over 116 moving features
Kemmeren 2014	1,437	0.238	−0.005	0.348
Messner 2023	4,279	0.179	0.008	0.278
Hu 2007	259	0.114	−0.052	0.235
Sameith 2015 Sm	81	−0.014	−0.077	−0.085

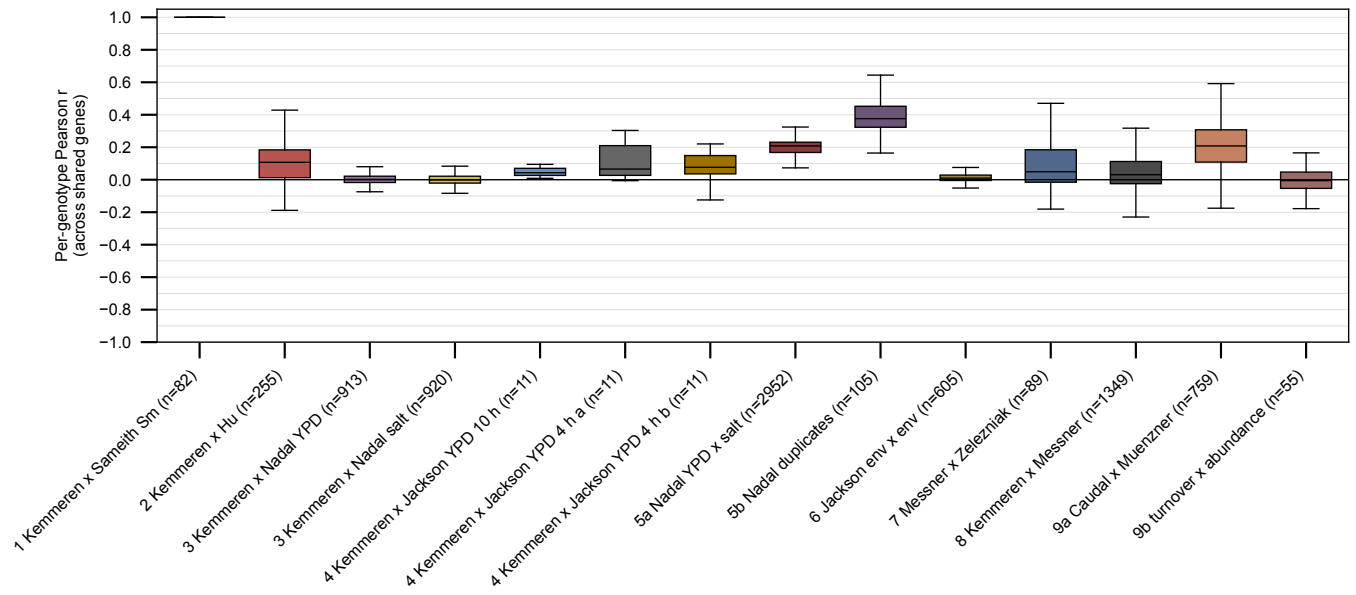


Figure 3: Agreement between datasets is low except for the Sameith copy. Per shared genotype, Pearson across the shared genes for every pair of Table 7; boxes are quartiles, whiskers 1.5 IQR, outliers not drawn.

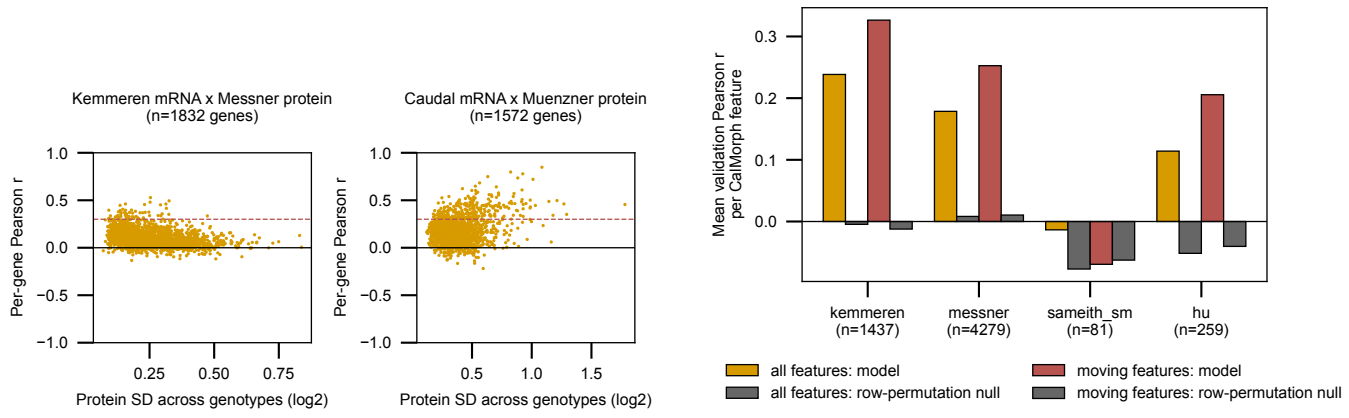


Figure 4: mRNA against protein, and expression to morphology. Left two panels: per-gene Pearson between mRNA and protein across the shared genotypes, against the protein's SD across those genotypes, for 1,832 genes over the deletions of Kemmeren and Messner and 1,572 genes over the isolates of Caudal and Muenzner; the dashed line marks $r = 0.3$. Right: mean out-of-fold Pearson per CalMorph feature of a ridge from each input matrix (Table 8), with its row-permutation null.

4 Which genotypes carry which labels ✗

A multitask model learns a shared representation only where one genotype carries more than one label. The census keys a genotype by its perturbed gene names, an isolate by its strain and a segregant by cross and identifier, and counts the genotypes each label and each pair of labels reach (Tables 9 and 10).

Table 9: Genotypes per label. Distinct genotype keys over every dataset carrying the label; Vaishnav is not keyed by gene and is left out.

Label	Genotypes, n
umi_count	82,479
protein_abundance	5,279
calmorph	4,678
expression_log2_ratio	3,685
expression_tpm	943
expression_score	262
degradation_rate	55

Table 10: Co-location. Genotypes carrying both labels, in any datasets. A label name can span datasets: `expression_log2_ratio` is Kemmeren, Sameith and Nadal-Ribelles; `umi_count` is Jackson, Boocock and N'Guessan; `protein_abundance` is Messner, Zelezniak and Muenzner.

Labels carried by the same genotype	Genotypes, n
expression_log2_ratio × calmorph	3,473
protein_abundance × calmorph	4,287
expression_log2_ratio × protein_abundance	3,317
expression_tpm × protein_abundance	759
umi_count × calmorph	12
expression_score × calmorph	259
isolates: Caudal × Muenzner protein	759
isolates: all three isolate datasets	55

Deletions. Morphology is the label most often shared. Of the 4,678 CalMorph genotypes, 4,287 also carry a protein profile and 3,473 an expression log ratio; the expression-by-protein overlap is 3,317 genotypes. That count includes Nadal-Ribelles, whose within-dataset agreement is near its null (Sec. 3.1); Kemmeren alone shares 1,349 genotypes with Messner. Kemmeren reaches Ohya on 1,437 genotypes and Messner on 4,279, the populations of the morphology ridge (Table 8).

Isolates. The isolate datasets are nested: all 55 turnover isolates are in the protein set and in Caudal's transcriptomes, and 759 of the 795 protein isolates have a Caudal transcriptome. No isolate carries a CalMorph profile.

Segregants and the rest. No segregant shares a key with any other dataset, and Boocock and N'Guessan share no segregant with each other (0 shared identifiers; Sec. 3.1 tests genotype identity rather than names). The 12 `umi_count` genotypes that carry morphology are Jackson's 11 transcription-factor deletions and the HO control.

5 The plan: every gene-level readout trained together ✗

Every gene-level readout of the store trains in one model, and the question it answers is an engineering one: does the model predict the transcriptome of a deletion it has not seen, and of a double deletion from its two measured singles? Table 11 gives the sets. An earlier version of this section proposed two separate training batches, deletions and then isolates with segregants; the owner superseded it on 2026-10-19, and the review pass of 2026-10-20 replaced the separate split draws by one fold file and put a linear ladder and a decision rule ahead of any training run.

Training. Twelve datasets train together: every dataset of the store except Sameith 2015 Sm, whose 82 profiles are Kemmeren's own arrays (82 of 82 identical), Sameith 2015 Dm, which is split B, and Ohya 2005, which is the probe. Vaishnav 2022 stays in the plan as the promoter signal: one sort-seq level per promoter insert, read at the record level through the reporter token of Sec. 7.2, whose promoter block is the frozen FUDT upstream encoder on the construct window; its linear evaluations run first and its joint entry waits on the Addgene 127546 sequence. Expression and protein readouts come off the perturbed graph: a gene's value is read from that gene's own perturbed node state, and the CLS state and the perturbed-gene summary z_S are the record-level sites (Sec. 8.1). The low-agreement datasets train too, and their measured agreement is what the arm without them reads: Nadal-Ribelles agrees with itself across batches at a median r of 0.043 against 0.022 for different genotypes and with Kemmeren at 0.002; two Jackson 2020 constructions of one genotype agree at 0.129 against 0.095; Hu 2007 released no replicate and agrees with Kemmeren at 0.107 per genotype, on an X score that is not a log ratio; Zelezniak 2018 agrees with Messner at 0.049 per genotype over 715 proteins though its own ceiling is 0.916; Muenzner 2024 turnover agrees with abundance at -0.006 per isolate.

Split A, deletion genes. One fold file holds every deletion gene's fold; fold 0 is held out of every deletion dataset at once, so a held-out gene is never seen as a perturbation on any readout, and the same genes test Kemmeren, Messner, Hu, Nadal-Ribelles and Zelezniak alone and jointly. Kemmeren holds out about 140 of its 1,400 eligible genes (the 82 Sameith singles are pinned to training). The score is the median per-gene Pearson across those strains; the per-strain Pearson is reported centered, because two different Kemmeren mutants already agree at a raw 0.284 (Sec. 3.1). The bound is Kemmeren's own-culture ceiling, 0.729 over 114 mutants with two cultures; the bars are the frozen-token ridge and kNN of Sec. 8.5 on the same genes.

Table 11: The sets of the plan. Records count the raw store (full pass of `data_shapes_043.py`, slurm job 5902). The fold file is not yet written, so a split's count is its expected fold-0 share under the stratified draw: a tenth of each deletion dataset's genes, a fifth of the isolates by strain name, of each segregant panel and of the promoter variants. Splits A, C, D and E are carved out of training records; split B is a dataset of its own and never trains.

Set	Datasets	Records or held-out units, n (expected)	Role
Training, every gene-level readout	Kemmeren 2014, Hu 2007, Jackson 2020, Nadal-Ribelles 2025, Messner 2023, Zelezniak 2018, Caudal 2024, Muenzner 2024 protein, Muenzner 2024 turnover, Boocock 2025, N'Guessan 2025, Vaishnav 2022	168,785	trained together; fold 0 of each key space is carved out of these records, fold 1 validates
Split A, deletion genes	every deletion dataset	140	gene fold 0: about a tenth of Kemmeren's 1,400 genes outside the 82 pinned Sameith singles; also Messner 2023 about 448, Nadal-Ribelles 2025 about 306, Hu 2007 about 26, Zelezniak 2018 about 10, Ohya 2005 about 468
Split B, Sameith doubles	Sameith 2015 Dm	72	every double; scored against the per-gene mean, the stronger single, the additive null and the fitted combination of its singles
Split C, isolates	Caudal 2024, Muenzner 2024 protein and turnover	196	isolate fold 0 of 979 strains: about 159 Muenzner and 11 turnover isolates, out of every readout
Split D, segregants	Boocock 2025, N'Guessan 2025	16,493	fold 0 within each cross: about 15,903 Boocock and 590 N'Guessan records
Split E, promoters	Vaishnav 2022	14,232	fold 0 of 71,158 promoter variants; linear evaluations first
Probe	Ohya 2005	4,678	ridge from the frozen latent to the CalMorph features, raw and growth-residualized; never trained
Dropped	Sameith 2015 Sm	82	the same arrays as Kemmeren (82 of 82)

Split B, the Sameith doubles. All 72 Sameith 2015 double deletions are held out and scored, centered, against their measured profiles and against four rungs that need no model: the per-gene mean, the stronger single, the additive null of the two Kemmeren singles (the paper's own expected value) and a fitted combination of the two singles. Sameith's Methods defer to Kemmeren's, so the doubles were profiled by the same procedure; no double has a replicate, and the Kemmeren ceiling of 0.729 is the bound for the same arrays.

Split C, the isolates. Fold 0 of the isolate folds, about 196 of the 979 Caudal and Muenzner strains with about 159 Muenzner and 11 turnover isolates among them, is held out of every readout of those isolates. The bar is kinship ridge (GBLUP) on the same isolates; the reference is how far another readout of the same isolate agrees, Caudal mRNA against Muenzner protein at a median per-isolate r of 0.208, against 0.031 for Kemmeren mRNA against Messner protein on deletions. Splits D (segregants, within cross) and E (promoters) come from the same file.

The morphology probe. Ohya 2005 CalMorph enters no training. Its 4,678 genotypes are run through the frozen model and a ridge reads the 116 moving features from h, z_S, $\bar{H}$ and their concatenation, raw and with the Costanzo single-mutant fitness regressed out, since growth matrices alone predict morphology nearly as well as the proteome (019 screen). A ridge on the measured inputs already reaches a median r of 0.348 from Kemmeren expression (1,437 shared genotypes), 0.278 from Messner protein (4,279) and 0.235 from Hu (259) over the 116 moving features, against row-permutation nulls near zero, and Ohya's own ceiling is 0.611 over 122 wild-type replicates on all 281 (Table 8). Sec. 8.6 gives the bars and the two reads.

The genotypes the model has not yet read. Isolates and segregants need a gene-level genotype the `Perturbation` processor does not build: marking every isolate perturbation sets 5,575 of 6,607 nodes, and a segregant sets none. Sec. 7 gives the replacement: each isolate as its own reference context built from its allele embeddings, with an interim arm that marks loss of function and absence on S288C, and the definite rule for segregants (a gene inherits the non-reference parent only inside a block called at posterior 0.9 or above), with the parent's real allele token. The posterior cut removes little on the pooled panels (a median of 0% of markers for Boocock pooled, 1.7% for N'Guessan) and a quarter to a third on Boocock's de novo cells, whose single cells carry medians of 2,000 to 4,000 UMIs; whether those cells train is decision P1. The measured genotype signal across segregant panels is small: nearest-genotype deviation r 0.113 against 0.070 for random pairs (Sec. 3.1).

5.1 What the three case studies teach ✗

Three earlier campaigns trained the same cell graph transformer on other labels, and each found a bar, a scoring trap and a protocol that the first 043 design did not carry.

Betaxanthin, experiment 031. The GO recipe with a rank term reached a fold-average held-out Spearman of 0.229 to 0.230 over six seeds (fold sd 0.003 to 0.006), 0.264 on the pinned fold; the annotation-free cell (composite sequence token plus metabolic block) 0.108 on the fold average and 0.125 on the pin. The bars were ridge on the deleted gene's raw ProtT5 token, 0.082 on the pin, and forests on ProtT5 (0.150) and on GO plus pFBA (0.204) on the fold average. What made it honest: five inductive folds over deletion genes with the external pin as fold 0, every arm paired by seed against a named control, the rule of a gain of at least 0.02 with 5 of 6 seeds up (9 of 12 at twelve), and a fixed budget, since epoch selection on validation was null (−0.009). The operator audit (jobs 4917, 4918) found the single-deletion score to be a learned affine read of the deleted gene's own token, reproduced by a ridge from that row: the inductive gap is in the representation, not the head. Split A is a single-deletion task and the same audit applies, so the first number to produce is the ridge from the deleted gene's frozen composite token to its profile; the first 043 design ran no such bar and proposed three initializations by two split seeds with no fold structure, where 031 found the pin's sampling sd (0.041) equal to the seed sd.

Environment, experiments 035 and 038. Nested ridge on raw FCFP4 counts reached a median centered Spearman per held-out compound of 0.303 over 123 compound-evaluations (035, slurm 3034) and 0.359 over 96 on the second build (038, slurm 3374); the transformer trained to saturation tied ridge on the first build (every entry point's compound-level interval through zero) and sat below it on the second (0.312 and 0.297; slurm 3700, 3701). The protocol that fixed the verdict: fit on the whole non-test pool as ridge does, a fixed 50 epochs keeping the last, three seeds, paired against ridge on the same held-out units, and the interval resampled over the 41 compounds rather than the 123 evaluations. Its centering lesson: each side is centered by its own training mean, since centering the prediction by the measured mean rewarded an untrained model at 0.356 after 60 steps (slurm 3009). The per-strain Pearson across genes has the same trap here: Kemmeren's different-mutant pairs agree at 0.284 (job 5893), the shared deletion response, which a model that emits the training-mean profile earns. Every per-strain Pearson is therefore reported centered, and the per-gene mean is the first rung.

Trigenic, experiments 010 and 030. The trigenic Pearson reached 0.454 ± 0.006 on the random split and fell to about 0.13 for every model on held-out query pairs, with the fitness-head model recovering 0.41 on one seed. On the random split the ladder read 0.390 for the query pair alone, 0.400 for the additive per-gene ridge, 0.425 to 0.427 for an embedding MLP and 0.438 to 0.455 for the transformer; on the 030 pool a zero-parameter lower-order null read 0.577 and 0.566 and boosted trees over the lower-order terms 0.743 and 0.700, above the joint transformer's 0.562 and 0.391 (job 3902). The per-entry dataset token (a 15-way one-hot projected to 8 dimensions at the readout) converged at a validation interaction Pearson of 0.5655 on one seed. Ordering generalized and magnitudes did not (held-out calibration slope 0.24, job 1605), and the slope is a property of the population scored (0.88 on the whole test split against 1.59 inside the predicted top 100). Split B carries the additive null but needs the whole ladder of rungs that see only a double's components, and because the binned head's point prediction is an expectation that shrinks toward the mean, Spearman sits beside Pearson and the calibration slope is named by population.

The unified protocol. Four steps, in order. (1) The linear ladder on CPU from the frozen embeddings, on the same held-out units the transformer will use (Sec. 8.5); nothing trains on a card until it is on file. (2) The rule, committed as `conf/decision\_rule.yaml` before the first GPU run: primary statistic per split, the control arm, a gain of at least 0.02, at least 5 of 6 paired seeds up, and a 95% interval resampled over held-out strains (genes for per-gene statistics) that excludes zero. (3) Training with the budget and window declared at launch; the validation fold monitors the escape and never selects the epoch; the test fold is read once. (4) Paired comparison by initialization seed, each arm against its named control, at the unit of generalization: held-out deletion gene, isolate, double, segregant or promoter, with both intervals computed and the unit-level one quoted.

Decisions. Table 12 collects every choice the design leaves to the owner, with two options, the measured fact that bears on it and the reconciled proposal with its named alternatives. The I rows come from the input side (Sec. 7.10), the O rows from the output side (Sec. 8.7), the P rows from putting the plan together, the S rows from the protocol the review pass added, D1 from the data per model part (Sec. 6.1) and E1 from the isolate embedding plan (Sec. 7.5).

Table 12: Owner decisions, one checklist. Each row is one choice the design cannot make alone. Option 1 is the reconciled main option; the proposed column states the decision and names its alternatives; "open" marks a choice with no proposal. The measured fact is the one its section cites. The last column is the section that carries the full argument.

Id	Decision	Option 1	Option 2	Measured fact	Proposed, with alternatives	Sec.
I1	Base gene token	composite frozen sequence token: FUDT upstream 1,003 bp and downstream 300 bp (768 each), ProtT5-XL (1,024), CaLM (768); per-block LayerNorm and projection, summed, plus the position term	learned identity table (the 019 model)	030 composite, trigenic, validation interaction Pearson 0.5655 at convergence (one seed); 031 betaxanthin composite 0.219 against ProtT5 alone 0.216; a one-hot row is not a function of sequence, so an unseen allele, accessory ORF or heterologous gene has no token	composite; the identity table survives only as the transductive-ceiling ablation and enters no claim. Alternatives: nucleotide-transformer windows in place of FUDT (hosts with no FUDT species token; one yeast arm), ESM2 650M in place of ProtT5 (cost only)	7.2
I2	Absence and deletion	two change types	one change type	126 natural absences per Caudal record (median); the similarity is measurable after training as the cosine of the two embeddings	two; report the cosine after training	7.3
I3	Isolates	own reference context: token matrix from the isolate's allele embeddings, encoded once and cached, 979 contexts; the operator carries only engineered edits (the HO deletion)	perturbation of S288C: loss-of-function plus absence marks, $M \approx 518$, allele tokens on the variant rows	a Caudal record carries 5,662 perturbations (median), 5,069 of them sequence variants; HO deletion on 116 of 943 Caudal records	option 1 is the main design; option 2 is the interim arm for a first run if the owner keeps the S288C route (owner question 1)	7.4
I4	Segregant gene membership	whole gene inside a block of posterior 0.9 or more, with the parent's real allele token when the parent is a collection member	gene midpoint inside a block	5 of 5 non-BY parents carry a Peter id in the loaders that is one of the 1,011 isolates of the gene-keyed allele tarball; straddling genes per segregant not counted	whole gene, parent allele tokens from the tarball (owner question 4 confirms)	7.3
I5	Chromosomal position	on for every class: reference coordinates, chromosome embedding scaled by chromosome copy number, Fourier features of the relative midpoint, the gene's pangenome copy number	off, or on only when segregants enter	Peter 2018: 342 aneuploidies in 193 isolates (19.1%), no translocation fraction reported; assemblies short-read, median N50 136 kb, 3,259 contigs, no per-isolate GFF	on for every class; isolate-own coordinates deferred to chromosome-scale assemblies; conserved gene order is a hypothesis	7.6
I6	Context tokens	split by kind: allele parent on the gene token; environment and experiment type read by the head; dataset only as the per-dataset bias at the head	every context as an extra key of the operator (the prototype)	a dataset key the trunk can read lets the latent learn dataset-specific states, and the morphology probe reads that latent for genotypes with no dataset token	option 1; dataset identity never enters the trunk or the operator in any arm	7.7
I7	Operator normalization at large M	softmax, magnitude fixed	sigmoid gate, magnitude grows with M	operator A run at $M = 1$ to 3 only; sigmoid gate 0.200 against 0.210, one seed	moot under I3 main, where M is 2 or less outside the segregants; softmax stays; the gate only if a class with M of 10 or more remains	7.8
I8	Accessory ORFs and the Vaishnav reporter	extra off-graph tokens from the same composite encoders	report and skip	78 accessory presences per Caudal record (median), 1,614 distinct pangenome ORFs; Vaishnav 71,158 records, one reporter (YFP)	option 1; the reporter's promoter block is the frozen FUDT upstream encoder on the construct window, which needs Addgene 127546 in the constructs tier (owner question 2)	7.2
O1	Heads	one head per readout level (gene-level, record-level), shared across readout types and conditioned by the type token; targets standardized per dataset; per-dataset learned bias	six heads by readout type	same-type agreement 0.002 to 0.107 (expression), 0.049 (protein); an affine map cannot raise a correlation; no joint training scored	option 1; the six-heads-by-type form is the control arm	8.2
O2	Experiment-type token	ontology path with a modeling-side grouping (expression, protein, promoter, fitness, interaction, metabolite, morphology)	flat class id	the served ontology is flat, 34 leaves under phenotype	path token (the owner's tokenizing of the ontology); flat id as the ablation	8.3
O3	Where the tokens enter	at the head: type, dataset and environment tokens concatenated to the gene state, with the per-dataset bias	type and environment attended in one layer after the operator	030 dim-8 per-entry readout token: validation interaction Pearson 0.5655 at convergence, one seed; 035 attended token +0.008 against ridge, CI through zero	head as main; the attended layer as arm 1 (the form a compound-structured environment will need); no token as the ablation; dataset never in the trunk	8.3
O4	Loss	HL-Gauss binned prediction on standardized targets, K = 32 over $[-4, 4]$ z, sigma 0.75 bin widths, expectation as the point	19-quantile pinball; MSE	019 morphology on z_S: MSE escaped 12 of 12 runs, pinball 8 of 10; no binned head has run here	bins as main; pinball as arm 1 (the owner's preference over MSE); one MSE cell as the reference; every gain a hypothesis	8.4
O5	Dataset weighting	equal per dataset	record count; agreement weighting	Boocock holds 52% of the gene-level values (records times the median keys on the index)	equal; agreement weighting dropped; record count as an arm; P1 decides whether the de novo cells train	8.4

Id	Decision	Option 1	Option 2	Measured fact	Proposed, with alternatives	Sec.
O6	Nadal-Ribelles sentinels	$\lvert \log_2 \mathrm{FC} \rvert > 20$ set to missing before standardization	keep the stored value	20.2% of cells are sentinels	missing, before the standardization statistics are fitted	8.4
O7	Jackson target	deletion effect against the HO control of the same environment	absolute $\log_2(\mathrm{CPM}+1)$ with the environment token	construction pairs agree at 0.129 against 0.095 on the effect; one genotype across environments at 0.010	the effect; Jackson not scored alone	8.6
O8	Split A leakage rule	held-out gene out of every deletion dataset, from the global fold file	out of the expression datasets only	Kemmeren different-mutant floor 0.284 per strain; the 82 Sameith singles must stay in training for split B	every deletion dataset, by the fold file (S1)	8.6
O9	Split C	fold 0 of the 5-fold isolate partition	a fifth of the Muenzner isolates alone	979 Caudal and Muenzner strains, 759 shared, 55 with turnover; Caudal against Muenzner 0.208 per isolate	fold 0, about 196 isolates (about 159 Muenzner, about 11 turnover)	8.6
O10	Budget and window	declared with the launch; alone and joint matched on optimizer steps	epochs, with validation selecting the epoch	019's 1,000-epoch runs scored lower over 900 to 999 than over 400 to 499 in 7 of 8; 031 epoch selection on validation null (-0.009)	option 1; validation never selects the epoch; test read once	8.6
P1	Boocock de novo cells	train all de novo single cells under the posterior rule	train the pooled segregants only	de novo cells: median 2,000 to 4,000 UMIs, 24 to 35% of markers below posterior 0.9; Boocock is 52% of gene-level values	open; the every-readout direction includes them	5
P2	Split A against split B	pin the 82 Sameith singles to the training folds	draw split A from all 1,482 Kemmeren genes	split B's additive null needs both singles of each double in training	pin the singles (kept)	5
S1	Global split file	one file, `conf/splits/global_split_v1.json` (pydantic `GlobalSplit043`, seed 0, store sha256): genes 10-fold, isolates 5-fold, segregants 5-fold within cross, promoters 5-fold; fold 0 test, fold 1 validation	separate draws per split	alone and joint runs must read the same held-out records; Kemmeren keeps 1,400 eligible genes after pinning	one file; an alone run is the joint config with one dataset listed; a clade-disjoint isolate file as the hard split (owner question 6)	8.6
S2	Decision rule	gain of 0.02 or more, 5 of 6 paired seeds up, 95% interval over held-out strains excluding zero	seed count alone	019 seed sd 0.006 (expression) and 0.008 (proteome); 031 pin sampling sd 0.041 equaled the seed sd	option 1, committed before the first GPU run; 9 of 12 when fold 1 runs as a second split	8.6
S3	Linear evaluations first	CPU ladder from the frozen tokens before any card is used	train first	031 operator audit: the single-deletion score is an affine read of the deleted gene's own token, ridge on raw ProtT5 0.082; the token ridge has never been measured on expression	option 1: per-gene mean, ridge and kNN, GBLUP for isolates, marker ridge for segregants, k-mer ridge for Vaishnav, the four-rung double ladder	8.5
S4	Per-strain statistic	centered on the training-mean profile, raw beside it	raw only	Kemmeren different-mutant pairs agree at 0.284 per strain (job 5893)	centered, with the floor quoted beside any raw number (owner question 7)	8.6
S5	Transfer statistic	joint minus alone per dataset, paired by seed, strain-resampled interval	joint score alone	alone runs see fewer records per epoch, so equal epochs are not equal budgets	option 1, matched on optimizer steps	8.6
S6	Morphology probe control	ridge from frozen h, z_S, mean H and their concatenation, raw and growth-residualized; fold-0-gene subgroup	raw probe only	measured-input ridge bars Kemmeren 0.348, Messner 0.278, Hu 0.235 (job 5892); growth matrices predict morphology nearly as well as the proteome (019 screen)	option 1; the 019 head 0.149 is the inductive bar	8.6
S7	Reporting	Spearman beside Pearson; calibration slope named by population	Pearson only	010 calibration slope 0.88 on the whole test split, 1.59 inside the predicted top 100	option 1, for every head	8.6
D1	Data per model part	count the data each part trains on before the run (Table 14)	count after	18,476 deletion events against 9,026,112 variant events; 775,856 distinct protein alleles over 6,081,165 allele records	option 1, regenerated with the store	6.1
E1	Isolate embedding plan	dedup by sequence hash, four encoders, CPU alignment first, GPU when the cards free	embed every (isolate, gene) pair	775,856 distinct proteins and 1,198,539 distinct CDS over 1,011 isolates (`count_distinct_alleles.py`)	option 1; ProtT5 about 2.7 h on two cards under stated throughput assumptions (hypothesis), ESM2 a third of that (owner question 3)	7.5

Questions for the owner.

1. **Isolates as reference contexts or the interim arm.** The main design encodes each isolate as its own reference context (979 cached contexts, the operator carrying only engineered edits) and needs the per-context encoder; the interim arm marks loss of function and absence on S288C with allele tokens and needs nothing new. Which runs first?

2. **Addgene 127546.** May its plasmid sequence be fetched into `torchcell-constructs/` with a provenance record, so the Vaishnav construct gets the 1,003 bp window the gene promoter encoder expects?

3. **ProtT5 or ESM2 for the first isolate pass.** 775,856 distinct proteins: about 2.7 hours of ProtT5 on two cards under the stated throughput assumptions, ESM2 about a third of that.

4. **Segregant parents.** The five non-BY parents carry Peter ids that are all members of the gene-keyed allele tarball; confirm that their real allele tokens replace the parent id on marked genes.

5. **Headline joint arm.** With or without Hu, Jackson and Nadal-Ribelles (Kemmeren agreement 0.107, 0.043 to 0.076, 0.002)? They can only lower the Kemmeren score if their signal does not transfer; the rule names its control before the first read.

6. **Clade-disjoint isolate split.** A second isolate fold file, disjoint by Peter 2018 clade, beside the random one, once the clade is on the isolate descriptor?

7. **The quoted per-strain statistic.** Confirm that the centered per-strain Pearson, with the raw value and the 0.284 floor beside it, is the form quoted.

6 Data as tensors ✗

Figure 5 follows a record from the store to a batch, and Table 13 gives each dataset's shape from a full pass over the 173,617 records. A record is a pair of experiment and experiment reference; its genotype becomes a set of marked genes on the 6,607-gene index G of `gene_set.json`, its phenotype a vector y over G with `NaN` where a gene is not measured, and its environment, dataset and experiment type become integer tokens.

What the dataset module returns. Per sample, `dataset_043.py` returns the marked genes as indices on G (`gene_index`), a change type per marked gene (`pert_type`), the allele parent per marked gene for the segregants (`allele_parent`), the promoter sequence and scaffold for Vaishnav, the environment, dataset and experiment-type ids, the label scale (`y_scale`), the label $y \in \mathbb{R}^{6,607}$, one scalar for Vaishnav and the 281 CalMorph features for Ohya. The collate function concatenates the marked genes with a batch vector and stacks y into a $B \times 6{,}607$ tensor with its measured mask. A mixed batch of one record per dataset, 15 rows, carries 21,728 marked genes, 21,713 of them from the five isolate and segregant rows (the six-record debugger run).

Three shapes that bear on the design. The isolate and segregant rows are dense: a Caudal isolate marks 4,952 to 5,995 of 6,607 genes and a segregant 1,415 to 3,158 under the definite rule, against one for a deletion, so the perturbation operator sees three orders of magnitude of marked-set size across datasets (counts on the debugger records). The definite rule leaves gaps: on the six debugger records 697 to 2,693 genes fall between blocks and 157 to 704 under a block below posterior 0.9. Two label-key maps run at read time: Nadal-Ribelles keys are standard names (1,441 of 6,796 on G by raw name, 6,325 after the map) and Messner keys are UniProt accessions (0 of 1,850 by raw name, 1,850 after). A key with no position on G is counted per sample and dropped from y, never guessed.

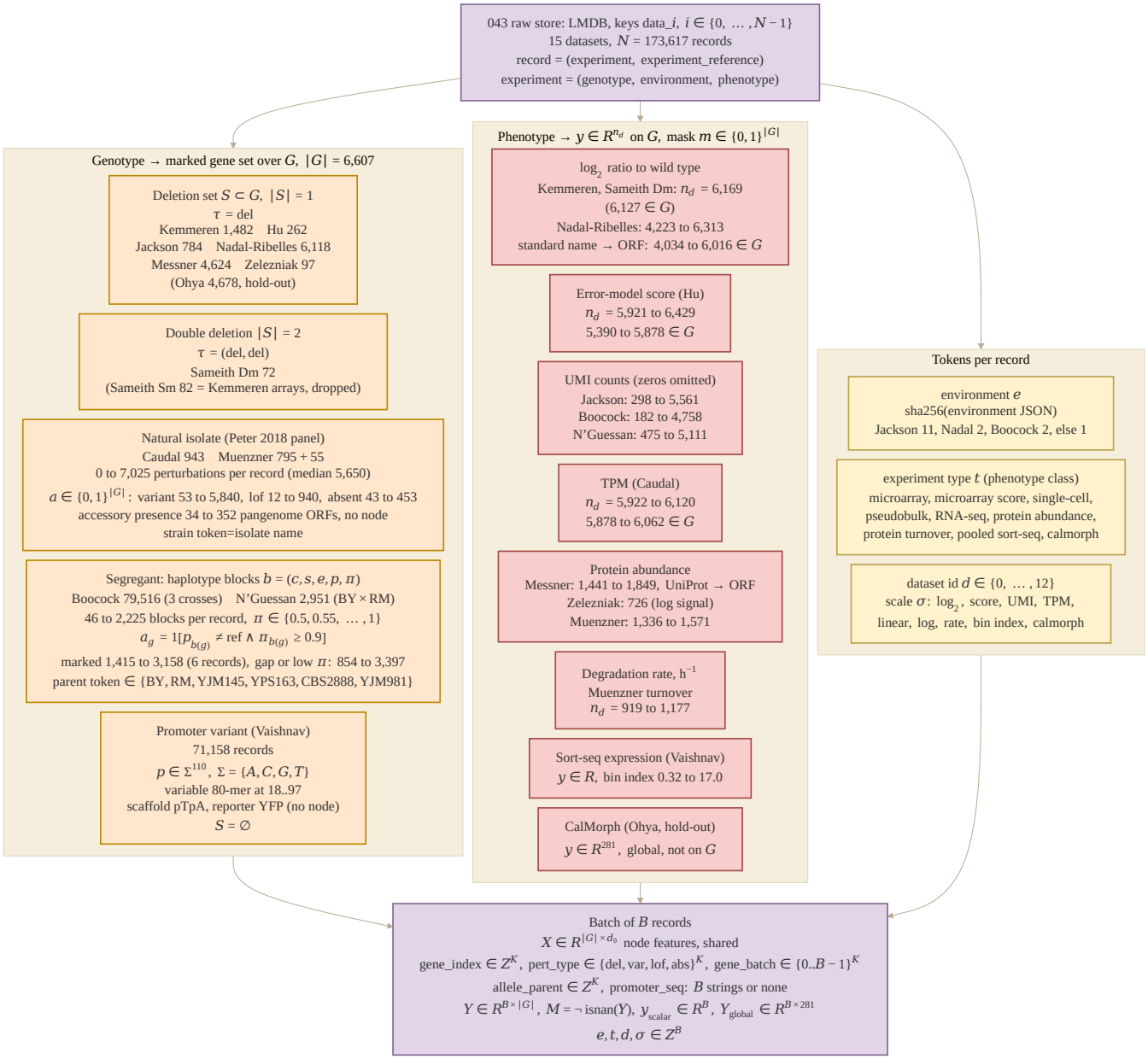


Figure 5: A record of the store becomes a marked gene set, a masked label vector and four tokens. Left, the five genotype classes and the counts each marks; middle, the nine released label scales with the number of label keys per record and how many of them fall on G; right, the environment, experiment-type and dataset tokens; bottom, the batch the collate function returns. Rendered 212 mm wide and placed at 180 mm, a scale of 0.85.

Table 13: Every dataset of the store as tensors. One row per dataset, from a full pass over the raw store (`data_shapes_043.py`, slurm job 5902). Perturbations per record count the released perturbation list (a segregant has none; its genotype is a list of haplotype blocks). Label values are the finite minimum and maximum over a sample of each dataset's values, on the released scale. Label keys on G count the keys that land on the 6,607-gene index after the read-time maps. Where a cell shows one number, minimum, median and maximum are equal; otherwise it reads median (minimum to maximum).

Dataset	Role in the plan	Records, n	Genotype class (records)	Perturbations per record	Label key, graph level	Label values, min to max	Label keys per record	Label keys on G per record	Environments, n
Kemmeren 2014	train; split A, gene fold 0 out	1,482	Genotype	1	`expression_log2_ratio`, node	−6.72 to 6.17	6,169	6,127	1
Sameith 2015 Sm	dropped: Kemmeren's arrays	82	Genotype	1	`expression_log2_ratio`, node	−4.31 to 5.23	6,169	6,127	1
Sameith 2015 Dm	split B: every double held out	72	Genotype	2	`expression_log2_ratio`, node	−6.25 to 4.8	6,169	6,127	1
Hu 2007	train; gene fold 0 out	262	Genotype	1	`expression_score`, node	−8.35 to 8.05	6,381 (5,921 to 6,429)	5,832 (5,390 to 5,878)	1
Jackson 2020	train; gene fold 0 out	784	Genotype	1	`umi_count`, node	1 to 21,655	4,028 (298 to 5,561)	4,009 (298 to 5,501)	11
Nadal-Ribelles 2025	train; gene fold 0 out	6,118	Genotype	1	`expression_log2_ratio`, node	−33 to 32.9	5,866 (4,223 to 6,313)	5,665 (4,034 to 6,016)	2
Messner 2023	train; gene fold 0 out	4,624	Genotype	1	`protein_abundance`, node	0.0313 to 393,221	1,819 (1,441 to 1,849)	1,819 (1,441 to 1,849)	1
Zelezniak 2018	train; gene fold 0 out	97	Genotype	1	`protein_abundance`, node	1.62 to 11.2	726	726	1
Ohya 2005	probe only, frozen latent	4,678	Genotype	1	`calmorph`, global	0 to 56,974	281	none, features not genes	1
Caudal 2024	train; split C, isolate fold 0 out	943	Genotype + natural isolate	5,662 (195 to 7,025)	`expression_tpm`, node	0 to 88,829	6,050 (5,922 to 6,120)	6,006 (5,878 to 6,062)	1
Muenzner 2024 protein	train; split C, isolate fold 0 out	795	Genotype + natural isolate (794); Genotype (1)	5,628 (0 to 7,025)	`protein_abundance`, node	0.0906 to 181,277	1,540 (1,336 to 1,571)	1,540 (1,336 to 1,571)	1
Muenzner 2024 turnover	train; split C, isolate fold 0 out	55	Genotype + natural isolate	5,662 (4,903 to 6,430)	`degradation_rate`, node	0.0169 to 2.44	1,055 (919 to 1,177)	1,055 (919 to 1,177)	1
Boocock 2025	train; split D, fold 0 within cross out	79,516	SegregantGenotype	0	`umi_count`, node	1 to 2,301	962 (182 to 4,758)	962 (182 to 4,758)	2
N'Guessan 2025	train; split D, fold 0 out	2,951	SegregantGenotype	0	`umi_count`, node	1 to 12,339	2,592 (475 to 5,111)	2,591 (475 to 5,109)	1
Vaishnav 2022	train once the plasmid window exists; split E, fold 0 out	71,158	Genotype	1	`expression_level`, node	0.321 to 17	1	none, one scalar	1
Total		173,617						6,607 genes on the index	

6.1 Data per model part ✗

Each part of the model trains on a different slice of the store, and Table 14 counts it before any run. Two readings follow from it. First, the operator's training signal for engineered edits is 18,476 deletion events over at most 6,607 genes, dominated by Nadal-Ribelles (6,118 records, agreement with Kemmeren 0.002) and Messner (4,624, proteome); the clean expression deletions are Kemmeren's 1,482. The nine million variant events are natural variation, whose home under the main isolate design is the reference context (Sec. 7.4) rather than the operator. Second, the gene-level value count is dominated by Boocock's single cells, 52% of the training values at a median of 962 labeled genes per cell, so Boocock sets the gradient volume unless the equal per-dataset weighting of Sec. 8.4 or decision P1 limits it. The distinct allele counts set the isolate embedding bill: 775,856 proteins, not 6.1 million (Sec. 7.5). Segregant marks under the definite rule are not in the census; on the six debugger records a segregant marks 1,415 to 3,158 genes, so hypothesis (untested): the 82,467 segregant records carry on the order of 10^8 marks in all.

Table 14: The data that trains each part of the model. Records, events and genotypes from the full shape pass (`data_shapes_043.py`, slurm job 5902) and the store census; isolates counted by strain name. Gene-level values are records times the per-record median of label keys on the 6,607-gene index, so they are approximate. An event is one stored perturbation; deletion events include the isolates' HO deletions, the Ohya probe and both Sameith sets. Allele counts are one pass over the Peter 2018 gene-keyed tarball over all 1,011 isolates (`count_distinct_alleles.py`); a protein allele is the translation of a distinct CDS truncated to whole codons.

Part of the model	Trained by	Records or events, n	Genotypes or reference contexts, n	Gene-level values, n (records times median keys on the index)	Distinct genes or alleles, n
Gene tokens (W_k, T, P) and the gene-level head	the 11 gene-level training datasets	97,627	93,648 genotypes	1.5×10^8; Boocock 2025 52%, Nadal-Ribelles 2025 23%, Kemmeren 2014 6%	at least 6,383 labeled genes (Jackson 2020); union not computed
Isolate reference contexts (I3 main) or isolate strain tokens	Caudal 2024, Muenzner 2024 protein and turnover	1,793	979 isolates by strain name (943 Caudal, 795 Muenzner, 759 shared)	6.9×10^6	label keys on the index per record (median): Caudal 2024 6,006, Muenzner 2024 protein 1,540, Muenzner 2024 turnover 1,055
Segregant genotypes (parent allele tokens)	Boocock 2025, N'Guessan 2025	82,467	Boocock crosses B 44,784, A 28,137, 3004 6,595; N'Guessan 2,951; 6 distinct parents	8.4×10^7	label keys on the index per record (median): Boocock 2025 962, N'Guessan 2025 2,591
Reference encoder, distinct reference cells	all		today 1 (S288C); under I3 main 979 isolates plus 5 non-BY parents, 984 contexts, each encoded once		S288C index 6,607; 6,015 reference genes per isolate in the allele tarball
Operator: deletion	deletion sets and the isolates' HO deletions	18,476 events			distinct deleted genes: Sameith 2015 Dm 96; Kemmeren 2014 1,482; Hu 2007 262; Nadal-Ribelles 2025 3,059; Messner 2023 4,476; Zelezniak 2018 97; Ohya 2005 4,678; Sameith 2015 Sm 82; Jackson 2020 12
Operator or reference context: variant	isolates	9,026,112 events	1,793 records		genes with a variant: Caudal 5,973; 6,081,165 allele records over 1,011 isolates and 6,015 genes, 1,198,539 distinct CDS, 775,856 distinct proteins; per gene median 188 CDS, 109 proteins, max 991 and 967
Operator or reference context: loss of function	isolates	682,189 events			genes: Caudal 4,137, Muenzner 3,952
Operator or reference context: absent	isolates	239,636 events			genes: Caudal 1,111, Muenzner 1,105
Extra tokens: present accessory ORF	isolates	150,913 events			pangenome ORFs off the index: Caudal 1,614, Muenzner 1,643
Extra token: promoter reporter	Vaishnav 2022	71,158 events	71,158 promoter variants	71,054 scalars, none on the index	1 reporter (YFP)

7 The input side: sequence, genotype and the perturbed cell graph ✗

Five genotype classes have to enter one model so that every gene-level readout trains together: engineered deletions and double deletions, natural isolates, segregants and promoter variants, and the same input has to hold a gene the training set never saw: an isolate allele, an accessory ORF, a heterologous cassette, a gene of another host. That constraint fixes the abstraction between the stored genotype and the perturbation operator: a **gene token** per gene from a frozen composite sequence encoder, never from an identity table; a **reference context** per distinct genome (S288C, each natural isolate, each segregant parent), encoded once and cached; a **change type** per gene per strain that says how the strain's copy differs from its reference; a **position** term carrying chromosome, relative coordinate and

copy number; and **context tokens** for what is true of the record rather than of a gene, which the heads read (Sec. 8.3). The perturbation operator then reads the strain's engineered edits exactly as it reads deleted genes today. Everything measured below is read off the 043 store census, the shape pass, the genomes tier or the stored embedding tables; what is proposed is marked as such, and the decisions collected in Sec. 7.10 carry one recommended option each. Table 15 defines the symbols and Fig. 6 draws the path.

Table 15: Symbols of the input side, alphabetical. Widths are those of the 1,084,619-parameter expression model of experiment 019 (`conf/cgt_expr_012.yaml`: $d = 90$, $h = 9$ heads) and of the stored S288C embedding tables.

Symbol	Meaning
$a_i \in \mathcal{A}$	allele parent of gene i in a segregant: the parent whose block covers the gene; $\varnothing$ for every other gene and every other genotype class
B	strains in a batch
$c_i \in \mathcal{C}$	change type of gene i under the strain's genotype (Table 16); ref when the strain carries the reference copy
$\mathcal{C}$	the change-type vocabulary, seven members
d	width of a gene token and of a gene state, 90
$E \in \mathbb{R}^{(\lvert G\rvert+1)\times d}$	encoder output over the reference cell graph: row 0 the CLS state h, rows 1 to $\lvert G\rvert$ the wild-type gene states h_i
$f_k : \Sigma^* \to \mathbb{R}^{d_k}$	the four frozen sequence encoders of the composite token, $k \in \{\text{up}, \text{down}, \text{prot}, \text{cds}\}$: FUDT upstream (768), FUDT downstream (768), ProtT5 (1,024), CaLM (768)
$G^{(s)}$	the gene set of reference context s; $\lvert G^{(\text{S288C})}\rvert = 6{,}607$, written G where the context is S288C
$H \in \mathbb{R}^{B\times\lvert G\rvert\times d}$	perturbed gene states, the operator's output (`H_genes_pert`)
h, $h^{\text{pert}} \in \mathbb{R}^d$	the CLS state before and after the operator
M	marked genes of a strain, $M = \sum_i m_i$; the operator's key count
$m \in \{0,1\}^{\lvert G\rvert}$	the mark: $m_i = \mathbf{1}[c_i \neq \text{ref}]$
$P_i \in \mathbb{R}^d$	the position term of gene i (Eq. 3)
$\mathbf{p}_t \in \mathbb{R}^d$	token of the t-th marked gene, the operator's key and value source
$\pi_i = (\chi_i, \rho_i, \kappa_i)$	position of gene i: chromosome index $\chi_i \in \{0, \dots, 16\}$ (0 the mitochondrion), relative reference coordinate $\rho_i = \frac{1}{2}(\text{start}_i + \text{end}_i)/L_{\chi_i}$, pangenome copy number κ_i
$s_i(\gamma) \in \Sigma^*$	realized sequence of gene i under genotype γ
$T \in \mathbb{R}^{\lvert\mathcal{C}\rvert\times d}$, $A \in \mathbb{R}^{\lvert\mathcal{A}\rvert\times d}$	learned change-type and allele-parent embeddings; $T_{\text{ref}} = A_\varnothing = 0$
t_{type}, t_{env}	the ontology-path and environment tokens, read by the heads (Sec. 8.3); the dataset enters only as the per-dataset bias at the head
$X^{(s)} \in \mathbb{R}^{\lvert G^{(s)}\rvert\times d}$	gene tokens of reference context s, computed once and cached; $X = X^{(\text{S288C})}$
$\tilde{X} \in \mathbb{R}^{B\times\lvert G\rvert\times d}$	realized gene tokens of a batch, $\tilde{X}_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}}$
$z_S \in \mathbb{R}^d$	pool over a strain's marked rows of H (`pooled_perturbed`)

7.1 What enters the model today ✗

The 019 expression model builds its gene tokens from a learned table, not from sequence: `conf/train_cgt_multitask.yaml` sets `node_embeddings: []` and `learnable_embedding.enabled:true`, so X is an `nn.Embedding(6607,d)` indexed by gene order, and the CLS row is one learned vector shared by every strain. The encoder runs once per step on the wild-type graph; a strain enters only through the operator, whose token for a deleted gene is that gene's encoder state, $\mathbf{p}_t = h_{g(t)}$. The Methods write $\mathbf{p}_t = r(e_t) + \mathbf{t}_{\tau_t} + m_t\mathbf{w}$ with a perturbation-type embedding $\mathbf{t}_\tau$; the code implements neither the type term nor the magnitude term (perturbation-operator document, Table of symbols). Three consequences for the 043 classes are measured:

- A deletion and any other mark on the same gene produce the same key and the same value, since the token is the gene state alone.
- The `Perturbation` processor marks every perturbation of a record, so a Caudal isolate enters with 5,575 of 6,607 genes in `pert_mask` (Sec. 2), and under operator A their 5,575 values are averaged into one vector delivered to every gene.
- A segregant enters with `pert_mask` all false and is the reference strain.

The sequence encoders exist on `main` and are already keyed by gene: `NodeEmbeddingBuilder` serves ProtT5 on the protein (`prot_t5_xl_uniref50`), nucleotide-transformer windows of 5,979 bp around the ORF and of 1,003 bp upstream and 300 bp downstream, the fungal up-down transformer on the upstream and downstream flanks, CaLM and codon frequency on the coding sequence, and one-hot identity. Each is one frozen row per reference gene, concatenated into `cell_graph["gene"].x` and projected by the model's preprocessor. None of them has a per-strain version.

7.2 The gene token ✗

Definition. Gene i's token is the composite of four frozen sequence encoders on its realized sequence under reference context s, each block standardized and projected, plus the position term of Sec. 7.6:

$$X_i^{(s)} = \sum_{k\in\{\text{up,down,prot,cds}\}} W_k \,\mathrm{LN}_k\big(f_k(s_i^k)\big) \;+\; P_i, \qquad W_k \in \mathbb{R}^{d\times d_k}, \tag{1}$$

with f_{up} the fungal up-down transformer (FUDT) on the 1,003 bp window ending at the start codon, f_{down} its downstream model on the 300 bp window from the stop codon, f_{prot} ProtT5-XL on the protein and f_{cds} CaLM on the codon sequence; the widths are those of the stored S288C tables (768, 768, 1,024, 768; `data/scerevisiae/\{fudt,protT5,calm\}\_embedding/processed`). Each block has its own layer normalization, so a 1,024-wide block does not outweigh a 768-wide one, and its own projection, so one block can be swapped without retraining the others. This is the composite of experiments 030 and 031 (`--node-embeddingsfudt\_upstreamfudt\_downstreamprot\_T5\_allcalm`); on betaxanthin it reads 0.219 against ProtT5 alone at 0.216 (031 rounds 232 and 233), and in the trigenic per-entry-token run it reaches a validation interaction Pearson of 0.5655 (seed 0, epochs 164 to 181, random split). The encoders are frozen: the 031 operator audit found the single-deletion score to be a learned affine read of the deleted gene's own sequence token, and nothing measured says fine-tuning a 1.2-billion-parameter encoder on 1,482 deletion profiles would help. The projections W_k, the change-type table T and the position term are the trained part.

A learned identity table is not an option for the base token: a one-hot row is not a function of sequence and cannot produce a token for an isolate allele, an accessory ORF, a heterologous cassette or a gene of another host. It survives as one ablation arm, the transductive ceiling of gene identity on splits A and B, which measures what the composite loses by not memorizing the gene and enters no claim.

Alternatives. Nucleotide-transformer windows (2,560 wide, five-prime 1,003 bp and three-prime 300 bp, both stored) in place of the two FUDT blocks, for a host with no FUDT species token; FUDT is a species-aware fungal model and covers no bacterium, so one 043 arm carries the swap and compares the two families on yeast before a host change forces it. ESM2 650M in place of ProtT5, on cost alone: on betaxanthin adding it read 0.216 against 0.219 (031 round 233), and it embeds at about a third of ProtT5-XL's cost, which matters for the isolate pass of Sec. 7.5.

A gene unseen in training. gets its token by the same four functions: protein from its CDS, the two windows from the flanks it sits between. For an isolate allele the CDS is a member of the gene-keyed tarball and the flanks come from the isolate's own assembly (Sec. 7.4); for an accessory ORF the CDS is in `allORFs\_pangenome.fasta.gz` (7,796 ORFs); for a heterologous cassette the flanks are the cassette's promoter and terminator; for *P. kudriavzevii* the tier already holds `proteins.faa`, `cds.fna`, the GFF3 and the FASTA.

Deleted gene: the row stays, marked. The deleted gene's row is neither removed from X nor zeroed. Removing it is the token-edit design, which costs one encoder pass per strain because the gene set then differs by strain (`notes-tex/perturbation-operator`, Sec. 4); the 019 configuration pays the encoder once per step and that is the dominant cost it was tuned around (`cgt_expr_012.yaml` header). Zeroing it before the encoder has the same cost and a second defect: a zero row through layer normalization is the normalized bias, one constant vector, so every deleted gene becomes the same token and the identity of what was deleted is lost before the first attention layer. The row therefore keeps its reference token and the deletion is carried by the change type, $c_i = \text{del}$, which the operator reads as a learned vector added to the token (Sec. 7.8). The gene's own readout entry is then predicted from a token that says "this gene, deleted", which is the information the deleted-gene flag $\delta_{i,t}$ of the perturbation-operator document supplies; the predicted $+0.036$ against a measured -2.42 for that entry is what the flag, and now the mark, exists to fix.

Variant gene: the allele's own token. For a natural isolate every `SequenceVariantPerturbation` points at the isolate's allele through a sha256-pinned `sequence\_ref`, so f_k is computable on the allele for every one of the 9,026,112 variant perturbations of the store. Measured over all 1,011 isolates and the 6,015 gene files of the tarball: 6,081,165 allele records, 1,198,539 distinct CDS alleles (19.7%) and 775,856 distinct protein alleles (12.8%); the median gene has 188 distinct CDS and 109 distinct protein alleles, the most polymorphic 991 and 967 (`results/distinct\_alleles.json`, Table 14). The variant row is the composite on the allele plus the change type, $X_i \leftarrow \sum_k W_k\,\mathrm{LN}_k(f_k(s_i^{k,\text{strain}}))+P_i+T_{\text{var}}$, from the first run. The reference-token-plus-T_{var} form of the earlier draft would give 5,069 of an isolate's 6,607 genes one shared vector and leave its genotype to the 362 loss-of-function calls, 126 absences and 78 accessory presences (medians per Caudal record, `results/data\_shapes\_043.json`).

Loss of function and absence. A `NaturalLossOfFunctionPerturbation` is a released call on a gene (nonsense plus SIFT, or homozygous frameshift), 383 per Caudal record (mean); a `NaturalGeneAbsencePerturbation` is a reference gene the isolate lacks, 135 per record (mean). Both keep the reference row and carry their own change type. Most loss-of-function genes also carry a sequence variant; one gene carries one change type, resolved by precedence (Table 16).

Accessory gene present. A `NaturalGenePresencePerturbation` names a pangenome ORF with no row on the S288C index, 78 per Caudal record (median), 1,614 distinct. It gets an extra token from Eq. 1 on the pangenome CDS

and its assembly flanks, with T_{pres}, and its copy number from `genesMatrix\_CopyNumber.tab.gz` (1,011 isolates by 7,796 ORFs) in the position term. The prototype's report-and-skip (`off\_graph\_marks`) becomes report-and-encode. A heterologous cassette in a production strain is the same object with a designed promoter, so this is the row the first-named engineering manipulation needs.

Promoter variant. A Vaishnav record is one 80 bp variable region inside the constant `pTpA` scaffold (17 bp five-prime flank, 80 bp insert, 13 bp three-prime flank, 110 bp in all) driving YFP on the CEN plasmid Addgene 127546 in S288C::*ura3* in SC-Ura; its label is the read-weighted mean of the 18 sort bins of log(YFP/RFP) the sequence was observed in, one scalar, 71,054 values over 71,158 records (`vaishnav2022.py`; `results/data\_shapes\_043.json`). The inserts are designed, not native: random 80-mers, genetic-algorithm designs, simulated-evolution trajectories and single-mutant series of native 80-mers (design labels of the GEO table). Vaishnav stays in the plan. There is no second promoter encoder: the construct goes through the gene promoter encoder, the 1,003 bp window ending at the YFP start codon, built from the plasmid sequence with the insert substituted and embedded by the same frozen f_{up} every native promoter uses, and the reporter token is

$$u_{\text{rep}} = W_{\text{up}}\,\text{LN}\left(f_{\text{up}}(\text{window}_{1003}(p))\right) + W_{\text{prot}}\,\text{LN}\left(f_{\text{prot}}(\text{YFP})\right) + W_{\text{cds}}\,\text{LN}\left(f_{\text{cds}}(\text{YFP}_{\text{cds}})\right) + T_{\text{prom}}, \tag{2}$$

an extra off-graph token like an accessory ORF, read at the record level with the Vaishnav bias. A designed promoter and a native promoter then share one function, which is what a promoter-driven heterologous cassette needs later, and the 71,158 records train the projection W_{up} the native promoters also use.

Order of entry. The plasmid sequence of Addgene 127546 is not in `torchcell-constructs/` (which holds only `cachera2023\_cri\_spa\_plasmids`) and must be fetched with a provenance record before the window can be built (owner question 2). Linear evaluations on Vaishnav run first and need no plasmid for two of their three inputs: ridge on k-mer counts ($k \leq 6$), ridge on the one-hot 80-mer, and ridge on the frozen f_{up} window once the plasmid is in. Joint training entry is gated on the plasmid window, not dropped. Hypothesis (untested): ridge on the frozen window reaches $r > 0.5$ on the sort-seq level. Alternatives: a small CNN trained from scratch on the 80-mers, the paper's route but with 20 to 30 million random sequences against the 71,158 loaded here, which wins only if the frozen read shows no signal; or fine-tuning FUDT on Vaishnav, which wins if the frozen read is positive but far below the paper's reported performance and costs one FUDT forward per gene per step for the whole model.

7.3 The change-type vocabulary ✗

Table 16 is the vocabulary and the map from every `perturbation_type` of the store onto it, with the count of marks per record. The mask is $m_i = \mathbf{1}[c_i \neq \text{ref}]$, so what the operator sees as perturbed is decided here and not by the processor's rule of marking everything. A training run may restrict further, for example marking only loss of function and absence for isolates in the interim arm of Sec. 7.4; that is a mask over change types, $m_i = \mathbf{1}[c_i \in \mathcal{C}']$, and the count of distinct genes under each $\mathcal{C}'$ is not in the census yet (the 383 and 135 are counts of calls, and a gene can carry two calls).

Table 16: Change types, the store's perturbation types that map onto them, and marks per record. Counts are the census's perturbation counts divided by records (`results/store_census.json`); the isolate rows are means over records, the others exact. Precedence orders the types for a gene that carries several marks: the first in the order wins, so a gene that is gone outranks a call on its sequence, which outranks a present variant allele. Segregant marks are derived from haplotype blocks, not from a perturbation list.

Change type	Precedence	Perturbation types	Datasets	Marks per record
absent	1	`natural_gene_absence`	Caudal, Muenzner protein, Muenzner turnover	135, 133, 134
deletion	2	`kanmx_deletion`, `sga_kanmx_deletion`, `sga_natmx_deletion`, `barcoded_kanmx_deletion`, `marker_deletion`, `deletion`	Kemmeren, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak, Ohya, Sameith Sm (1 each); Sameith Dm (2); isolates (HO: 0.12, 0.11, 0.07)	1; 2; 0.12
loss of function	3	`natural_loss_of_function`	Caudal, Muenzner protein, Muenzner turnover	383, 377, 373
variant	4	`sequence_variant`; a segregant gene in a non-reference block	Caudal, Muenzner protein, Muenzner turnover; Boocock, N'Guessan	5,047, 5,017, 5,059; segregants not yet counted
present	5	`natural_gene_presence` (off graph)	Caudal, Muenzner protein, Muenzner turnover	84, 84, 90
promoter	6	`variable_promoter_reporter` (off graph)	Vaishnav	1
reference	implicit	none	every unmarked gene	$6{,}607 - M$

Segregants under the definite assumption. A `SegregantGenotype` is a cross, two parents pinned to assemblies and a list of `HaplotypeBlocks` ({chromosome, start, end, parent, posterior, n_markers} in reference coordinates). Gene i is marked $c_i = \text{var}$ with a_i the non-reference parent when one block of that parent with posterior ≥ 0.9 contains the whole gene, $\text{start}_b \leq \text{start}_i$ and $\text{end}_i \leq \text{end}_b$. A gene in a reference-parent block, in a block below the threshold,

straddling a block edge or in the unassigned gap between two blocks stays reference: the crossover is unresolved by construction and nothing is guessed about it. The threshold removes little on the pooled panels (the median marker fraction below 0.9 is 0.000 for Boocock pooled and 0.017 for N'Guessan, Table 6) and much on the de novo cells (0.236 to 0.352). The parent token names the parent as the loader does: BY and RM for the pooled panels; YJM145, YPS163, CBS2888 and YJM981 for Boocock's other crosses. The loaders write chromosomes as `chrI` to `chrXVI` and the genome keys them as integers, so the gene table maps once. The probabilistic form, $m_i \in [0, 1]$ equal to the covering block's posterior, is the later step and changes nothing above except that the mask becomes a soft weight on the mark's embedding.

The parent's real allele. The loaders pin each non-BY parent to a Peter 2018 strain id through `SegregantParent.peter_strain_id` (RM `AAA`, YJM145 `AAS`, YPS163 `ACD`, CBS2888 `ABL`, YJM981 `ADI`; `bloom2019.py PARENT_PETER_ID`), and all five ids are among the 1,011 isolates of the gene-keyed allele tarball (`count\_distinct\_alleles.py`, which records the tarball's isolate ids; checked in `primer\_043\_tables.py`). A marked gene therefore carries the parent's actual allele token, as an isolate's variant row does, with A_{a_i} beside it; owner question 4 confirms the route.

7.4 Natural genomes as reference contexts ✗

An isolate is not a small perturbation of S288C. Under the earlier draft a Caudal isolate entered the operator with about 5,000 keys and the batched operator scored $6{,}607 \times 5{,}047 \approx 3.3 \times 10^7$ pairs per head per strain; that is the operator doing the encoder's work. The formulation is to encode one reference cell and apply the operator to what was done to it, so in the main design each isolate is its own reference context: its token matrix $X^{(s)}$ is built from its allele tokens (Eq. 1 on every gene it carries, accessory ORFs included, absences omitted), the encoder runs on it once, the result is cached, and the operator carries only the isolate's engineered edits (an HO deletion, on 116 of 943 Caudal records). The census names 979 isolates by strain name (943 Caudal, 795 Muenzner protein, 759 shared; all 55 turnover isolates in both; the shape pass counts 794 distinct isolate descriptors in Muenzner's 795 records, 978 in all), so the encoder runs on 979 more contexts over a training run, not once per record. This is also the form a host with no S288C needs, so another host is a special case of it rather than a new code path.

The interim arm. If the owner keeps the perturbation-of-S288C route for a first run (owner question 1), the isolate enters on the S288C graph with loss-of-function plus absence marks ($M \approx 518$, sums of the Caudal means before precedence) and the allele token on every variant row. It needs no per-context encoder and is the arm the main design is compared against.

What the genomes tier holds. `peter2018\_1011\_assemblies`: the de novo assemblies (4.0 GB tarball, one contig-level FASTA per isolate, 1,010 members in the index), the gene-keyed allele tarball (6,015 reference genes, one CDS per isolate, with the gene's R64 coordinate in the header), the pangenome FASTA, the presence, copy-number and frameshift matrices, the loss-of-function workbook and Table S1 (ploidy, zygosity, aneuploidy, HO deletion). It holds no per-isolate GFF and no chromosome-scale assembly: "The resulting assemblies had a median N50 of 136 kb and a median number of contigs of 3,259" (Peter 2018, Methods). A gene's position on the isolate's own assembly is therefore (contig, offset, strand) after aligning its CDS to the assembly, which the feasibility run on main did for 40 genes of three isolates (40 of 40 mapped, identity 99.6 to 99.9%, full 1,000 bp upstream flank on 93.75 to 97.5%; `experiments/embeddings/results/poc\_isolate\_fudt\_feasibility.json`).

The isolate genome object. The missing object owns an isolate's sequence and coordinates on its own assembly: a pydantic `IsolateGenome` with the isolate id, its assembly member, the reference assembly set, the `NaturalIsolateDescriptor`, the per-chromosome copy number, one `IsolateGene` per gene (state, pinned CDS reference, CDS and protein hashes, pangenome copy number, R64 span, assembly locus or a typed gap) and the alignment's processing record. Built once per isolate from the alignment, written as a derived GFF3 beside a per-gene JSON under the assembly set's `derived/<isolate>/` with provenance, and loaded through `AnnotatedGenome`, it has the same interface as `SCerevisiaeGenome` (windows, CDS, protein), exports GenBank and GFF3 through the existing path, and is what "inherit this genome and engineer from it" instantiates: a `Genotype` whose reference is the isolate. Two additions follow: `ReferenceGenome` gains an `assembly\_ref`, and the model's gene index becomes per reference context.

7.5 The isolate embedding plan ✗

What exists. S288C rows for every composite member, 6,607 genes each, under `data/scerevisiae/\{fudt,protT5,calm,nucleotide\_transformer,esm2\}\_embedding/processed/`. No isolate embedding exists; `data/scerevisiae/caudal2024\_isolate\_embeddings/` holds only the feasibility run's BLAST caches. The runner on main, `experiments/embeddings/compute\_isolate\_embeddings.py`, extracts an isolate's alleles from the

tarball, translates them, embeds with ESM2, locates each CDS in the isolate assembly with BLAST+ and slices the 1,003 and 300 bp windows; it does not run ProtT5 or CaLM and writes one file per isolate per encoder.

What to compute. For the 979 isolates of the store and the five non-BY segregant parents, per reference gene: ProtT5 on the translated allele, CaLM on the CDS, FUDT upstream and downstream on the assembly-sliced windows; per accessory ORF present, the same four on the pangenome CDS and its assembly flanks. Every sequence is deduplicated by hash first, so an identical allele is one row: 775,856 distinct proteins and 1,198,539 distinct CDS over the whole collection (Table 14). Distinct flank windows need the alignment first and are not counted; hypothesis (untested): their dedup ratio is close to the CDS ratio.

Order of magnitude. Assumptions stated, nothing run: ProtT5-XL in fp16 at about 40 proteins per second per card for a mean yeast protein, FUDT and CaLM at about 400 sequences per second per card, BLAST+ at one to three minutes per isolate on one core. On two 48 GB RTX 6000 cards the 775,856 distinct proteins take about 2.7 hours of ProtT5 (ESM2 about a third of that), CaLM about 25 minutes, and each FUDT pass under an hour if the window dedup ratio matches the CDS ratio; a factor of three on the ProtT5 rate moves the stage between one hour and one day. Storage in fp16 by distinct allele is 1.6 GB for ProtT5, 1.8 GB for CaLM and under 2 GB per FUDT table, against 39 GB dense. Both cards are held by another session's 031 packs, so this is a plan; the CPU stages (allele extraction, BLAST+ alignment, the genome objects, the dedup index) need no card and run first.

Layout. `\$DATA\_ROOT/data/scerevisiae/isolate\_embeddings/peter2018\_1011/` holds a `manifest.json` (a pydantic `IsolateEmbeddingManifest`: encoders, versions, window geometry, input sha256s, dedup counts), one table per encoder under `alleles/` keyed by sequence sha256 (`protein.h5`, `cds\_calm.h5`, `window\_up\_1003.h5`, `window\_down\_300.h5`), one index `index/isolate\_gene.parquet` from (isolate, gene) to state, copy number, the four hashes, the flank lengths and the locus, and one `derived\_gff/<isolate>.gff3` per isolate. $X^{(s)}$ is the join of the index to the four tables, and the S288C reference is the same join with the reference hashes, so one code path serves both. The script is `scripts/compute\_isolate\_composite.py`, one slurm array over isolate chunks, with this configuration:

```
class IsolateCompositeConfig(BaseModel):
    model_config = ConfigDict(extra="forbid", frozen=True)
    isolates: tuple[str, ...]          # canonical ids, SACE_ prefix kept
    refgene_tarball: ArtifactRef       # allReferenceGenesWithSNPsAndIndelsInferred.tar.gz
    assembly_tarball: ArtifactRef      # 1011Assemblies.tar.gz
    pangenome_fasta: ArtifactRef       # allORFs_pangenome.fasta.gz
    copy_number_matrix: ArtifactRef    # genesMatrix_CopyNumber.tab.gz
    encoders: tuple[Literal["prot_t5_xl_uniref50", "calm",
                            "fudt_species_upstream", "fudt_species_downstream"], ...]
    window_up_bp: int = 1003
    window_down_bp: int = 300
    min_identity: float = 0.95         # below it the locus is a typed gap
    blastn: Path
    makeblastdb: Path
    out_root: Path
    device: str = "cuda"
    batch_size_protein: int = 16
    batch_size_window: int = 64
# stages, each idempotent and keyed by sha256 so a rerun skips done work:
#   1. extract_alleles   CPU  per isolate {gene: cds}; hash CDS and protein; append to index
#   2. align_and_slice   CPU  IsolateGeneLocus per gene; slice windows; hash; derived GFF3
#   3. embed_distinct    GPU  per encoder, the distinct sequences not yet in its table
#   4. verify            CPU  every (isolate, gene) resolves to a row or carries a gap
```

7.6 Chromosomal position ✗

Position is on for every genotype class. The term is

$$P_i = \kappa_{\chi_i} C_{\chi_i} + \mathrm{Fourier}_K(\rho_i)\, W_\rho + w_c \log_2 \kappa_i, \tag{3}$$

with $C \in \mathbb{R}^{17 \times d}$ the chromosome embedding (0 the mitochondrion) scaled by the chromosome's relative copy number κ_χ (1 for a euploid chromosome; from the descriptor's `ChromosomeCopyChange` over the basal ploidy), ρ_i the gene midpoint over the chromosome length in reference coordinates with K sine and cosine features, and κ_i the gene's pangenome copy number (1 for every S288C deletion set; a deletion is the change type, not a copy number of 0, so the two channels stay distinct). Rotary position in the attention layers is the alternative; it costs a code change and buys nothing until segregant linkage is shown to matter.

Why now. A frozen sequence token cannot separate paralogs with identical protein and near-identical flanks (the subtelomeric and tandem families); position is the only channel that does. A segregant's genotype is linkage, contiguous runs along a chromosome, and Fourier features make adjacency a smooth function the operator can read. Aneuploidy and copy-number variation are the one measured, isolate-varying position signal: "We identified a total of 342 cases

of aneuploidy that affected 193 isolates (19.1% of the 1,011 strains)", and "CNVs are heavily enriched in subtelomeric regions, whereas internal chromosomal regions are largely copy-number stable" (Peter 2018). The paper reports no fraction of isolates with translocations; the SI names major rearrangements in the Malaysian clade ($n = 6$) and one translocation flanking a Taiwanese introgression. Hypothesis (untested): gene order is conserved against S288C in nearly all of the 1,011, so reference coordinates are the right coordinates for every class until chromosome-scale isolate assemblies are in the tier; `IsolateGene.assembly_locus` holds the slot for them.

7.7 Context tokens ✗

What is true of a record rather than of one gene is of three kinds, and they go three places. The allele parent of a segregant is a property of each marked gene and enters the gene token as A_{a_i}. The environment (medium, temperature, compound) and the experiment type (the ontology path of the record's phenotype class) are read by the head: in the main design they are concatenated to the gene state at the readout, Eq. 13, the form of the 030 per-entry readout token; arm 1 lets the gene states attend to them in one layer after the operator, the 035 environment-encoder form, which a compound-structured environment will need (Sec. 8.3). The dataset identity enters nowhere before the readout, in any arm: it is the per-dataset bias at the head, which can fix a scale and offset and cannot fit the biology. A dataset token the operator could attend to would let the latent learn dataset-specific cell states, and the morphology probe reads that latent for genotypes that carry no dataset token. The prototype's `strain\_token\_vocab` therefore keeps `parent:*` only; `medium:*` and `dataset:*` leave the input side.

7.8 The perturbation operator with marks ✗

The operator is unchanged in form. For gene i and the strain's marked genes $t = 1, \dots, M$,

$$s_{i,t} = \frac{(h_i W_Q) \cdot (\mathbf{p}_t W_K)}{\sqrt{d_k}}, \qquad \Delta h_i = \sum_{t=1}^{M} \beta_{i,t}\, \mathbf{p}_t W_V, \qquad \hat{h}_i = \mathrm{LN}(h_i + \Delta h_i), \qquad h_i^{\mathrm{pert}} = \mathrm{LN}\big(\hat{h}_i + \mathrm{FFN}(\hat{h}_i)\big), \tag{4}$$

with $\beta_{i,t}$ the softmax over t (operator A), or the softmax over t and a null sink (operator C), as the perturbation-operator document defines them. What changes is the token:

$$\mathbf{p}_t = h_{g(t)} \quad \text{(code today)} \qquad \longrightarrow \qquad \mathbf{p}_t = h_{g(t)} + T_{c_{g(t)}} + A_{a_{g(t)}}, \tag{5}$$

which is the Methods' $\mathbf{t}_\tau$ term made concrete, with the allele parent beside it. At $T = A = 0$ it is the current operator exactly. Three things follow.

A variant and a deletion of the same gene are now different inputs. Under Eq. 5 the key and value of a marked gene depend on its change type, so the model can deliver a different vector to the genome for "*YFG1* deleted" and "*YFG1* carries the RM allele". Without it the two records are the same input with different labels, and the loss can only average them.

M stays small. With isolates as reference contexts (Sec. 7.4) the operator's key count is the number of engineered edits of the strain: 1 for the single-deletion sets, 2 for Sameith's doubles, 0 or 1 for an isolate, and for a segregant the marked genes of the definite rule (1,415 to 3,158 on the six debugger records) until segregants, too, are encoded as their own mosaic context, which the parent-allele tokens make possible. The softmax normalization of operator A is kept; the sigmoid gate is an arm only for a class with $M \geq 10$, which under this design is the segregants alone. The interim isolate arm marks loss of function plus absence ($M \approx 518$) with allele tokens on every variant row; the batched operator pads to the largest M in a batch, so that arm pads every deletion in a mixed batch to the isolate's width.

The self entry changes meaning. For a deletion, gene $g(t)$'s own readout entry is the transcript that is gone (mean −2.42 in Kemmeren). For a variant it is the allele's own expression, which may be up, down or unchanged. The deleted-gene flag $\delta_{i,t}\mathbf{e}_{\mathrm{self}}$ therefore has to carry the change type, $\delta_{i,t}\, \mathbf{e}_{\mathrm{self},c_{g(t)}}$, or the model is told every mark silences its gene.

7.9 Shapes end to end ✗

For a batch of B strains over the gene set $G^{(s)}$ of their reference context,

$$X^{(s)} = \textstyle\sum_k W_k \,\mathrm{LN}_k\left(f_k(s(\gamma_s))\right) + P \qquad \in \mathbb{R}^{|G^{(s)}|\times d}, \tag{6}$$

$$c \in \mathcal{C}^{B\times|G|}, \qquad m = \mathbf{1}[c \neq \mathrm{ref}] \in \{0,1\}^{B\times|G|}, \qquad a \in \mathcal{A}^{B\times|G|}, \tag{7}$$

$$\tilde{X}_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}} \qquad \in \mathbb{R}^{B\times|G|\times d}, \tag{8}$$

$$E = \mathrm{Enc}\left([\,x_{\mathrm{CLS}};\, X^{(s)}\,]\right) = [\,h;\, h_1;\ldots;h_{|G|}\,] \qquad \in \mathbb{R}^{(|G|+1)\times d}, \tag{9}$$

$$\{\mathbf{p}_{b,t}\}_{t\le M_b} = \{\,h_i + T_{c_{b,i}} + A_{a_{b,i}} :\; m_{b,i} = 1\,\}, \tag{10}$$

$$(h_b^{\mathrm{pert}},\, H_b) = \mathrm{Op}\left(h,\, \{h_i\},\, \{\mathbf{p}_{b,t}\}\right) \qquad \in \mathbb{R}^d \times \mathbb{R}^{|G|\times d}, \tag{11}$$

$$z_{S,b} = \mathrm{pool}\{H_{b,i} : m_{b,i} = 1\} \qquad \in \mathbb{R}^d. \tag{12}$$

The encoder in Eq. 9 runs once per reference context and is cached: once for S288C, as today, and once per isolate under the main design; the strain enters at Eq. 10. Eq. 8 is what the prototype returns as `X_strain`; the operator consumes its marked rows through `perturbation_indices` and `perturbation_indices_batch`, the layout `EquivariantPerturbationTransform.forward` already takes, with the type and parent terms added to the gathered keys. The readouts are unchanged in their sites: per-gene heads read H, the record-level head reads $[h^{\mathrm{pert}} \,\|\, \mathrm{mean}_i H_i \,\|\, z_S]$ (`GlobalHead(use_pert_summary)`), both with the context tokens of Sec. 8.3, and z_S for a single deletion is the deleted gene's own perturbed token.

The prototype, `torchcell/models/genotype_tokens_043.py`, implements Eqs. 7 and 8 over a single S288C context: a `ChangeType` enum with the map of Table 16, a pydantic `GenotypeTokenSet` built from deletions, from a perturbation list or from haplotype blocks under the definite assumption, a `SequenceEncoder` protocol with a precomputed-table implementation and a `PromoterEncoder` protocol with no implementation, and a `GenotypeEncoder` module returning X, $\tilde{X}$, c, m, a, the flattened mark indices and the padded strain tokens. Its test (`tests/torchcell/models/test_genotype_tokens_043.py`, 4 tests) builds a deletion, an isolate-style and a segregant-style token set over 12 genes and checks every shape, the mask counts $(1, 4, 5)$, the precedence of loss of function over variant, the parent rows and that an accessory ORF is reported rather than dropped. The prototype's `PrecomputedGeneEmbedding` is the S288C case of the allele-keyed store; an `AlleleTableEncoder` that joins the (isolate, gene) index to the per-encoder tables is the generalization, and `PromoterEncoder` is retired in favor of f_{up} on the construct window. The code follows these decisions in the next stage.

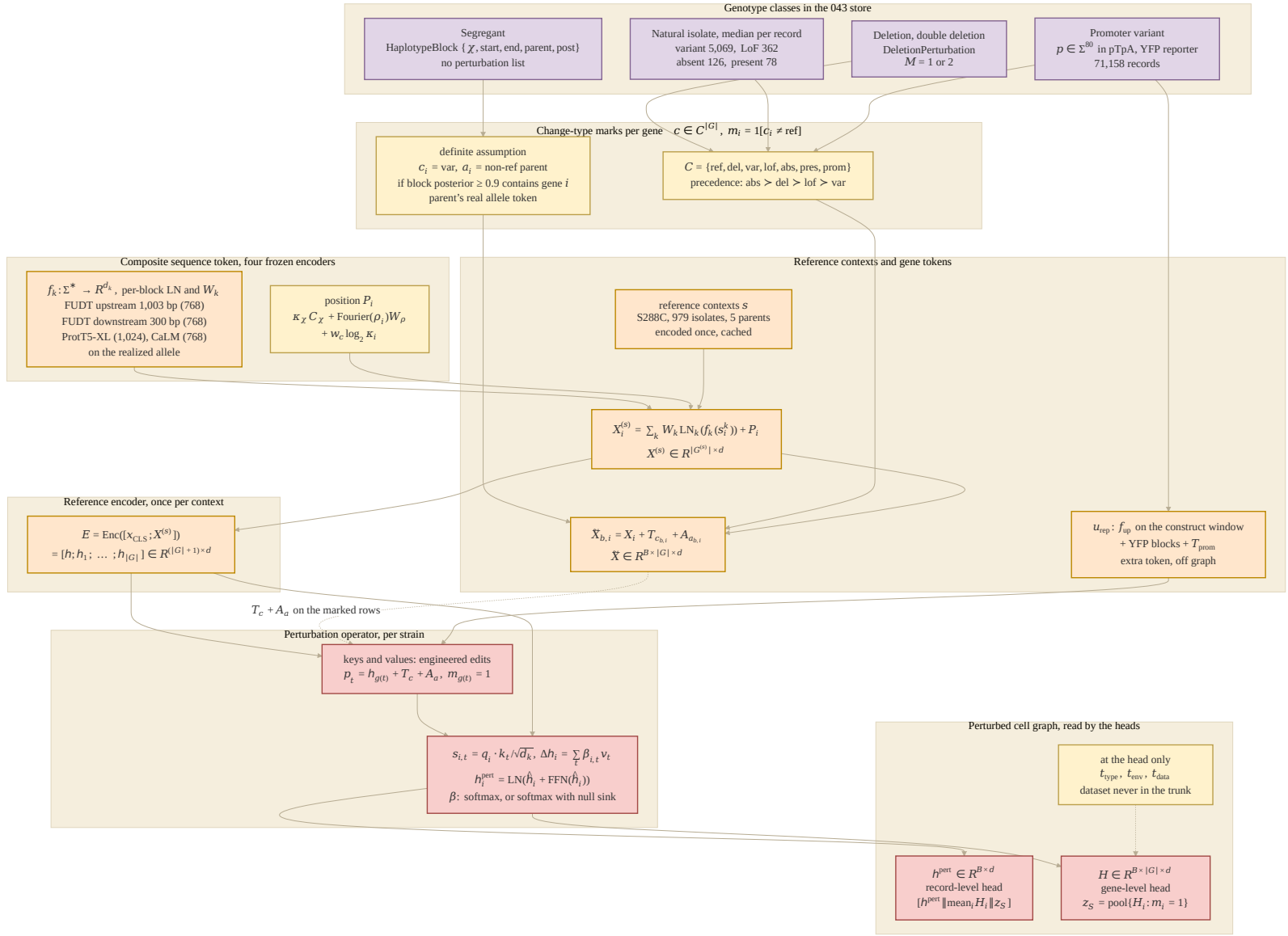


Figure 6: Every genotype class reaches the perturbation operator through one composite sequence token per gene and one encoder pass per reference context. Top, the genotype classes and their change-type marks; middle, the four frozen sequence encoders and the position term, the reference contexts (S288C, each isolate, each non-BY parent) and their gene tokens, the realized tokens and the reporter token, whose promoter block is the same FUDT upstream encoder on the construct window; the encoder runs once per context and is cached; bottom, the operator over the engineered edits and the perturbed states the heads read, where the type, environment and dataset tokens enter.

7.10 Owner decisions ✗

Each item names the recommended option first, then the alternative and the condition under which it would win. Table 12 carries the same rows with their measured facts.

1. **Base token.** Composite frozen sequence token, Eq. 1, with the position term. Alternatives: nucleotide-transformer windows in place of FUDT, for a host with no FUDT species token (one yeast arm compares the two families now); ESM2 650M in place of ProtT5 on cost. The identity table is the transductive-ceiling ablation only.
2. **Absence and deletion.** Two change types; report the cosine of T_{del} and T_{abs} after training.
3. **Isolates.** Own reference context, operator carries the engineered edits only (Sec. 7.4). Alternative: the interim loss-of-function plus absence mask with allele tokens, if a first run must precede the per-context encoder (owner question 1).
4. **Segregant gene membership.** Whole-gene containment, with the parent's actual allele token on marked genes; all five non-BY parents are collection members.
5. **Position.** On for every class, Eq. 3; isolate-own coordinates deferred to chromosome-scale assemblies.
6. **Context tokens.** Allele parent on the gene token; environment and ontology path read by the head (main) or attended after the operator (arm 1); dataset only as the per-dataset bias at the head, in every arm.
7. **Normalization at large M.** Moot under the main isolate design; softmax; the sigmoid gate as an arm for the segregant class alone.
8. **Accessory ORFs and the reporter.** Encode both as extra tokens from the same encoders; Vaishnav stays in the plan, its linear evaluations run first, and its joint entry waits on the Addgene 127546 sequence in the constructs tier (owner question 2).

8 The output side: readouts, heads, tokens, losses and the evaluation ✗

What follows is a design for the group to accept, change or reject. The owner's direction fixes three things: every gene-level readout is trained on; a readout is read off the perturbed graph, "from the node itself" for a gene-level value; and the question to answer is whether the model improves on held-out single deletions and on the Sameith double deletions, with the isolates as a further test and Ohya 2005 CalMorph as a probe of the learned state and never a training target. Everything else below is a choice with its facts, Fig. 7 draws the path, and Sec. 8.7 lists the choices the owner has to make.

8.1 Readout sites and the per-gene readout ✗

The trunk returns three tensors per record after the perturbation operator: the perturbed class token $h \in \mathbb{R}^d$, the perturbed gene states $H \in \mathbb{R}^{N \times d}$ over the $N = 6{,}607$ gene nodes, and the summary z_S. A gene-level value for gene g is read from that gene's own state,

$$\hat{z}_g = f_\theta\big([\, H_g \,;\, t\,]\big) + b_d, \qquad t = [\, t_{\text{type}} \,;\, t_{\text{data}} \,;\, t_{\text{env}} \,], \tag{13}$$

with f_θ one head shared by every gene, every dataset and every readout type of the gene level, t the record's tokens (Sec. 8.3), b_d the bias of dataset d and $\hat{z}_g$ the prediction on the dataset's standardized scale (Sec. 8.4); under the binned loss f_θ emits K logits and b_d is the vector c_d. The prediction for gene g depends on H_g and on nothing else of H; the strain reaches it through the trunk's attention and the perturbation operator, which is what "from the node itself" means in the equation. Four forms of f_θ are on the table.

- **Shared affine.** $f_\theta(x) = w^\top x + c$. The 019 head round measured the linear per-gene readout within noise of the two-layer one on Kemmeren expression (+0.002 over four seeds, `H_linear` against `H_ref`, `2-readouts.tex`), so the nonlinearity is not known to carry anything there.
- **Shared MLP with the per-dataset bias** (Eq. 13; the prototype `PerGeneReadout`).
- **Shared MLP with the strain vector concatenated.** 019's `H_concat` feeds $[H_g; c]$ with c the strain context, and was the one arm above the reference in every seed, +0.0135 with a seed sd of 0.0026 on the difference (`2-readouts.tex`, Table `tab:readouts-head`). In Eq. 13 this is t extended by z_S.

Table 17: Terms of this section, alphabetical. Terms of Table 1 are not repeated.

Term	Meaning
Additive null	For a double deletion xy and gene i, the single-mutant sum $M_{x,i} + M_{y,i}$, the expected value Sameith 2015 scores its epistasis against (Sec. 8.6). It needs no model.
Bias, per dataset	A learned offset applied to a shared head's output for every record of dataset d: a shift b_d under the pinball or MSE, a vector c_d over the bin logits under the binned loss. With targets standardized per dataset it is all that stays learned per dataset; it cannot change a correlation.
Centered per-strain Pearson	The Pearson across genes between a strain's prediction and its measurement after the training-mean profile of the dataset is subtracted from both. Kemmeren's culture 1 of one mutant against culture 2 of a different mutant agrees at a raw median of 0.284 (Sec. 3.1), the floor a raw per-strain Pearson carries.
Escape	A run whose training Pearson, in eval mode, passes 0.3. In 019 the genotype-only morphology runs sat on a plateau at 0.04 to 0.12 that an escape leaves within about 50 epochs (retrospective, `11-inductive.tex`).
Fold file	`conf/splits/global_split_v1.json`: one fold id per deletion gene, isolate, segregant record and promoter sequence; fold 0 is test, fold 1 validation; every run, alone or joint, reads it.
Head	The map from a readout site to a predicted value. Per level: one head for every gene-level readout and one for every record-level readout, conditioned by the type token. By type: one head per readout type. Split: one head per dataset.
HL-Gauss	The histogram loss of Farebrother et al. 2024: the target is a Gaussian over K bins, the loss its cross-entropy, the point prediction the expectation over bin centers, the uncertainty the bin entropy.
Perturbed-gene summary z_S	The sum of the perturbed states of the strain's perturbed genes, $z_S = \sum_{g \in S} H_g$; for a single deletion it is the deleted gene's own token. The input of the 019 `M_zs` readout and of the trigenic-interaction head.
Readout level	Gene-level (one value per gene of the index, read from H_g) or record-level (one value per record, read from h and z_S).
Readout site	A tensor of the perturbed graph a head reads: the perturbed CLS h, the summary z_S, or the gene states H.
Readout type	The physical quantity a head predicts: expression log ratio, absolute expression, expression score, protein abundance, protein turnover, promoter level.
Registered window	The epochs over which a run is scored, declared with the epoch budget before any run of the round is read; 019 used 500 epochs and epochs 400 to 499.
Token	A learned vector carried with a record and read by the head: the experiment type (ontology path), the dataset, and the environment.

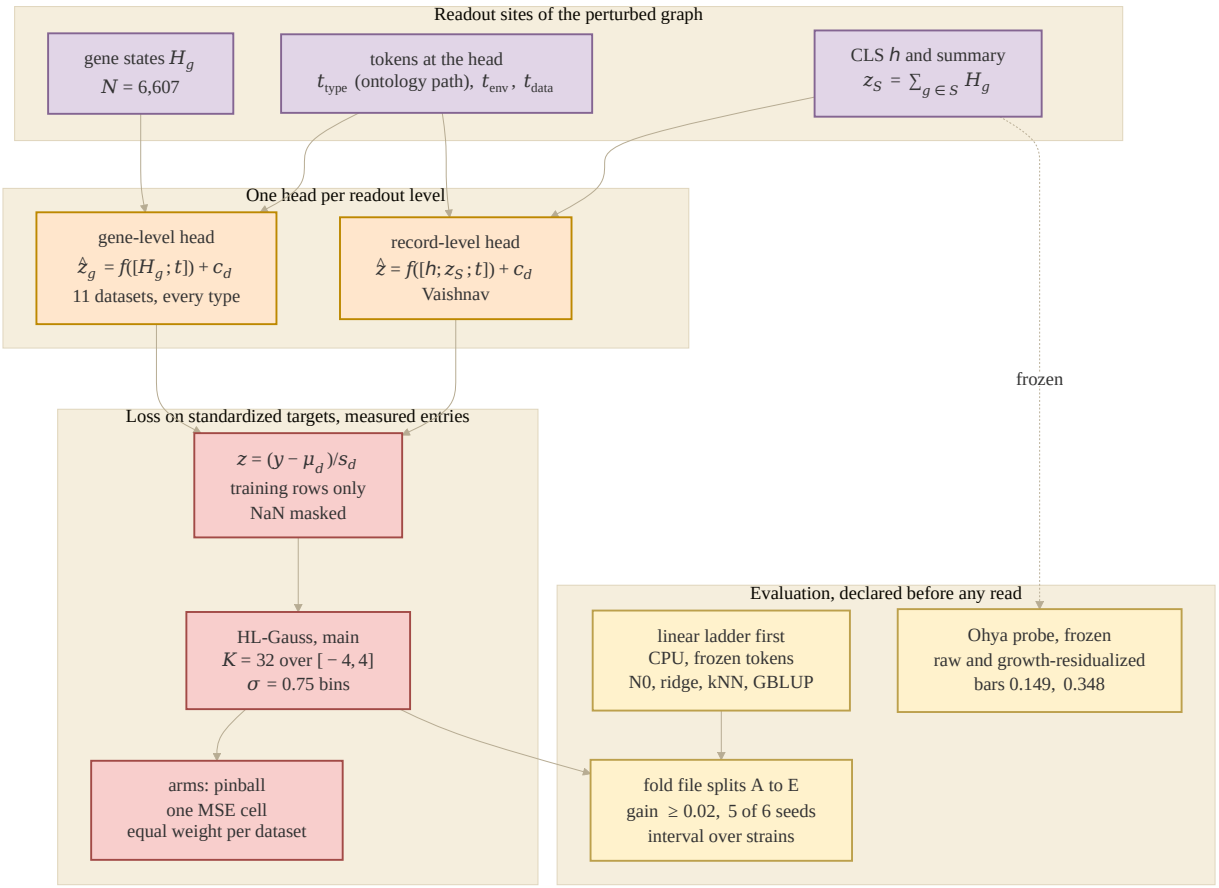


Figure 7: Gene-level readouts are read from each gene's own perturbed state by one head per readout level, conditioned at the head by the type, dataset and environment tokens, and Ohya is read only from the frozen latent. Top, the three readout sites of the perturbed graph and the three tokens; middle, the gene-level and record-level heads with the per-dataset bias, over targets standardized per dataset; then the binned loss over measured entries with its two arms; bottom, the evaluation: the linear ladder first, the held-out splits of the fold file and the morphology probe.

- **Cross-attention head.** One learned query per output feature over every token (019 `M_xattn`). On morphology it escaped under the pinball (4 of 4, 0.121 ± 0.029) and not under MSE (1 of 4) (`11-inductive.tex`). It reads the whole token set, so it is not a readout from the node itself; it is the alternative if the per-gene form stalls.

A record-level value (Vaishnav 2022, one level per promoter insert) is read from $[h; z_S; t]$ through an MLP, the `GlobalHead` with `use_pert_summary` that gave the 019 morphology result (12 of 12 escapes with MSE, 0.149 ± 0.022 over the registered window, `11-inductive.tex`). The plain CLS escaped 0 of 8 under MSE at two learning rates, so h alone is not a readout site this design uses for a trained target; it remains a probe input (Sec. 8.6).

8.2 Heads by level, by type or by dataset ✗

Two facts bound the choice. First, the datasets report different quantities on different scales (Table 18): a two-color $\log_2$ ratio to a wild-type pool, an error-model X score, a pseudobulk $\log_2$ fold change with sentinels at ± 23 to ± 33 in 20.2% of cells (028, `2-readouts.tex`), UMI counts, TPM, a linear MaxLFQ abundance, a log SWATH signal, a degradation rate per hour, a sort-seq bin level. Second, datasets of one quantity agree weakly where they share a genotype (Table 7): Kemmeren against Hu 0.107, against Jackson 0.043 to 0.076, against Nadal-Ribelles 0.002 per genotype; Messner against Zelezniak 0.049; Kemmeren mRNA against Messner protein 0.031. Standardization and a per-dataset bias reconcile the first fact and cannot reconcile the second: a head shared by two datasets that agree at 0.002 is asked to emit two unrelated vectors from the same H_g, told apart only by the tokens.

One head per readout level, the main design. Two heads: one gene-level head over every gene-level readout of the eleven gene-level training datasets, one record-level head over Vaishnav and any other scalar readout, each conditioned at the head by the type token of Sec. 8.3, with targets standardized per dataset and a learned bias per dataset. The type token carries what a by-type head would carry, so a readout type unseen in training lands on an existing path of the ontology and is scored by a head that already exists; a head per type cannot serve a type it never saw. What a shared head buys is the transfer of the readout function across datasets and types, which is the premise of the exercise and is unmeasured; what it risks is that a large low-agreement dataset pulls the shared function away from the one with a ceiling, which the equal per-dataset weighting of Sec. 8.4 limits.

Six heads by type, the control arm. Expression log ratio; absolute expression; expression score; protein abundance; protein turnover; promoter level. The datasets that would share a head and the agreement that supports it: Kemmeren and Sameith Dm (the same arrays and procedures; Sameith's Methods defer to Kemmeren, Sec. 3.2; the Kemmeren ceiling 0.729 is the only within-type ceiling) and, by scale alone, Nadal-Ribelles at 0.002; Jackson, Boocock, N'Guessan and Caudal under absolute expression (no shared genotype among them; Boocock and N'Guessan share no segregant, Sec. 3.1); Messner, Zelezniak and Muenzner under protein abundance (Messner against Zelezniak 0.049; Muenzner shares no genotype with either). The paired difference against the main design on the split-A score is the answer.

Split by dataset. Fifteen heads, no sharing below the trunk. Datasets then meet only in H, which is what every 019 round did with two array datasets (Kemmeren and Sameith) that are the same arrays. Nothing is known about the trunk's transfer between readout types from this campaign: proteome from genotype reached 0.085 ± 0.008 (v16, 12 runs) and expression 0.180 ± 0.006 (v12, 4 runs) in separate trainings (`10-data-review.tex`); a joint training of both was not scored on both. The alone runs of the transfer statistic (Sec. 8.6) measure what the split form would.

Table 18: Readout type and target transform per dataset. The transform column is what precedes the per-dataset standardization of Sec. 8.4; "as released" means the stored value. The transforms for Jackson, Caudal, Messner, Zelezniak and Muenzner are the ones the cross-dataset analysis used (Table 7); the Nadal-Ribelles and Vaishnav rows are proposals.

Dataset	Readout type	Released quantity, scale	Target transform
Kemmeren 2014; Sameith 2015 Dm	expression log ratio	two-color array, $\log_2$ (deletion / WT pool)	as released
Nadal-Ribelles 2025	expression log ratio	pseudobulk $\log_2$ fold change vs WT	as released, with $\lvert\cdot\rvert > 20$ set to missing (the zero-mean sentinel; 20.2% of cells)
Hu 2007	expression score	error-model X score vs BY4741, signed	as released; an X score is not a log ratio, and the type token says so
Jackson 2020	absolute expression	pseudobulk UMI counts, zeros omitted	$\log_2((\mathrm{CPM}+1)/(\mathrm{CPM}_{\mathrm{HO}}+1))$ against the HO deletion of the same environment
Boocock 2025; N'Guessan 2025	absolute expression	UMI counts per segregant	$\log_2(\mathrm{CPM}+1)$, centered per gene within the panel
Caudal 2024	absolute expression	RNA-seq TPM, linear	$\log_2(\mathrm{TPM}+1)$, centered per gene over the isolates
Messner 2023	protein abundance	SWATH MaxLFQ, linear	$\log_2$(strain / record reference)
Zelezniak 2018	protein abundance	SWATH log signal (SVA)	strain minus record reference
Muenzner 2024 protein	protein abundance	SWATH MaxLFQ precursor quantity, linear	$\log_2$, centered per protein over the isolates
Muenzner 2024 turnover	protein turnover	dynamic SILAC degradation rate, per hour	$\log_2$, centered per protein over the isolates
Vaishnav 2022	promoter level	sort-seq expression level, one scalar	standardized over the library

8.3 The experiment-type, dataset and environment tokens ✗

Three learned vectors travel with every record. The **experiment-type token** is the record's ontology phenotype class: `MicroarrayExpressionPhenotype` (Kemmeren, Sameith), `MicroarrayExpressionScorePhenotype` (Hu), `SingleCellExpressionPhenotype` (Jackson, Boocock, N'Guessan), `PseudobulkExpressionPhenotype` (Nadal-Ribelles), `RNASeqExpressionPhenotype` (Caudal), `ProteinAbundancePhenotype` (Messner, Zelezniak, Muenzner), `ProteinTurnoverPhenotype` (Muenzner) and `PooledSortSeqExpressionPhenotype` (Vaishnav), the classes of Table 2's rows. The **dataset token** is the dataset's id. The **environment token** is the record's medium, temperature and environmental perturbation, which separates Jackson's eleven conditions and Nadal-Ribelles' two.

The precedent. Experiment 035 gave the cell graph transformer a compound token z_E from a fingerprint MLP and asked whether it predicts Vanacloig 2022 for an unseen compound. Three entry points were measured on 123 compound-evaluations (`experiments.035`, rounds 9 and 10): a bilinear factor at the head (-0.007 paired against ridge, 95% CI -0.028 to $+0.015$), a gate on every gene's state (-0.025), and a token the genes attend to for one further layer ($+0.008$, CI -0.026 to $+0.045$; $+0.015$ against the head form, CI through zero). The ablation was the arm with no compound input (`cgt_genes`), whose centered score is zero by construction. Experiment 030 put a one-hot over its 15 source datasets, projected to 8 dimensions, at the readout of both heads; seed 0 on the random split converged at a validation interaction Pearson of 0.5655 by epochs 164 to 181 (W&B `xy3xnpau` and its continuations), and its encoder-side placement never ran.

Where the tokens enter, and the per-dataset bias. Three forms, for gene g of a record from dataset d with experiment type $\tau(d)$:

$$\text{(a)}\ \ \hat{z}_g = f(H_g)+b_d, \qquad \text{(b)}\ \ \hat{z}_g = f\big([\,H_g\,;\,t_{\tau(d)}\,;\,t_d\,;\,t_{\text{env}}\,]\big)+b_d, \qquad \text{(c)}\ \ H' = \text{Layer}\big([\,t_{\tau(d)};\,t_{\text{env}};\,H\,]\big),\ \ \hat{z}_g = f(H'_g)+b_d. \tag{14}$$

Duplicate encoding with a linear bias is the pairing in (b): the record's provenance is encoded once semantically, as the type and environment tokens shared across datasets, and once as free parameters, the dataset row and its bias, which absorb the instrument's scale and offset. With standardized targets the scale is fitted from the data and only the bias is learned; under the binned loss it is a bias vector $c_d \in \mathbb{R}^K$ on the logits, a learned prior histogram per dataset. Form (b) contains (a), since a one-hot t_d read by a head that is linear in it is a per-dataset bias.

The main design is (b), the 030 per-entry readout token's form: cheapest, measured to convergence on one seed, and it keeps every token out of the trunk. Arm 1 is (c), type and environment attended in one layer after the operator, the 035 environment-encoder form, the only one of 035's three with a positive paired mean against ridge, and the form a compound-structured environment will need, since a token the genes see can change which genes respond. The ablation is (a) with no token. The dataset id never enters the trunk in any arm, so the morphology probe reads a latent built without it. A new dataset is one new row of t_d and one new bias; a new dataset of a known type can be scored before training with t_d at its type's mean row and the bias at zero (hypothesis, untested, that this read is above the per-gene mean). The interface the head needs from the trunk is H, z_S, h and the three integer ids `collate_043` already emits; the isolate context and the change-type marks enter the operator and never the head, or the alone-against-joint transfer on the isolates cannot be attributed to the trunk.

Ontology path or flat id. A flat token is one row per class. A path token sums one row per node of the class's ontology path, so `MicroarrayExpressionPhenotype` and `RNASeqExpressionPhenotype` share an expression row and differ in their own (prototype `ExperimentTypeEmbedding`, `mode="dag"`). The served graph has no such path: every phenotype class is a direct child of `phenotype` in `biocypher/config/torchcell_schema_config.yaml` (the `phenotype member of` edge lists all 34 leaves), and in the pydantic schema only the two microarray classes share a parent below `Phenotype` (`TwoChannelMicroarrayExpressionPhenotype`). The main design writes the grouping on the modeling side: expression, protein, promoter, fitness, interaction, metabolite, morphology, with the served leaf below its group. This is the owner's tokenizing of the ontology; when the ontology unification lands a real path, `dag\_paths` reads it and nothing else changes. The flat id is the ablation, which says whether the shared group row carries anything (hypothesis, untested: with 15 datasets over 8 classes the type and dataset rows are nearly collinear, and the type row earns its place only on a dataset unseen in training).

8.4 Losses ✗

Each readout has one loss over the measured entries of its records only: a missing gene is `NaN` in the target tensor, contributes no value and no gradient, and a type with nothing measured in a batch returns a connected zero (prototype `MaskedTypedLoss`, the convention of `masked_mean` in `torchcell/losses/distributional.py`). Nadal-Ribelles sentinels are set missing first.

Standardization. Every target is standardized per dataset before the loss, $z = (y - \mu_d)/s_d$ with μ_d and s_d fitted on the training rows of dataset d, pooled over genes so a gene's deviation keeps its size. Every linear quantity is logged first

and every panel quantity centered per gene over the training isolates or segregants (Table 18). The learned per-dataset scale of the prototype is thereby fixed from the data; what stays learned per dataset is the bias of Eq. 14. The statistics are fitted on training records only and frozen for every linear evaluation, or they leak the held-out set's scale.

Binned prediction, the main design. The main loss is HL-Gauss (Farebrother et al. 2024): the standardized target is a Gaussian of width σ integrated over K bins and renormalized, the loss is the cross-entropy against that soft target, the point prediction is the expectation over bin centers, read back in released units as $\mu_d + s_d\,\mathbb{E}[z]$, and the bin entropy is the uncertainty. Bins are shared in z units: $K = 32$ equal bins over $[-4, 4]$, width 0.25, the two edge bins open, $\sigma = 0.75$ bin widths, so a target smears over about three bins. Vaishnav's released level is already a mean over 18 sort bins and is treated as a continuous scalar after standardization. The logits are $[B, 6{,}607, K]$, 6.8 million floats or 27 MB in fp32 at $B = 32$, against the 19-knot pinball's 16 MB; the attention over 6,607 tokens dominates either. The cross-entropy per entry is masked exactly as the MSE is; the soft target is built from the zero-filled safe target under the mask, so no `NaN` reaches the backward pass. Scalar heads (trigenic interaction, betaxanthin, the promoter level) use the same bins on their own standardized target. Pearson and Spearman are computed on $\mathbb{E}[z]$; accuracy within one bin and the mean entropy are reported beside them, the classification reading of the same prediction. Hypothesis (untested): the binned head escapes the marginal plateau at least as reliably as MSE on z_S, because its gradient at the marginal is the full mismatch of two histograms and not a shrinking residual. No binned head has run in this codebase, so every gain is a hypothesis until the arms below have their seeds.

Pinball and MSE, the arms. 019 trained expression with the 19-quantile pinball and read the median; on morphology, MSE on `M_zs` escaped 12 of 12 runs by epoch 410, the pinball on the same readout 8 of 10 and late (epochs 230 to 770), MSE on the CLS 0 of 8, CRPS 0 of 8 (`11-inductive.tex`). Nothing is measured on a per-gene head with any loss on these data. The pinball is arm 1, on the same seeds as the main design, and is the owner's stated preference over MSE. One MSE cell on the per-gene head is the reference: it is the only loss with a 12 of 12 escape on record (on z_S), so it tells a bins failure from a readout failure. The Pearson loss and ListMLE are not arms: five of six batch-32 Pearson runs collapsed to a constant output and the anchored ListMLE runs sat on the mean by epoch 250 (`2-readouts.tex`). A two-hot target is HL-Gauss at $\sigma \to 0$ and is covered by a σ sweep over 0.5, 0.75 and 1.0 bin widths if the main design is in doubt.

Weighting across datasets. By record the store is 173,617 samples, of which Boocock holds 79,516 and Vaishnav 71,158 (Table 2); by gene-level value Boocock's single cells hold 52% of the training values (Table 14); Kemmeren is 1,482 records. The main design weights each dataset's mean masked loss equally. Agreement weighting is dropped: it writes a judgment about which datasets are poor into the objective, and the arm without the low-agreement datasets (Sec. 8.6) answers the same question without it. Record-count weighting is the arm that shows what the two large panels do to the deletion score, and decision P1 settles whether Boocock's de novo cells train at all.

8.5 Linear evaluations first ✗

Nothing trains on a card before the ladder is on file. Every rung reads the fold file of Sec. 8.6, scores the same held-out units the transformer will, and runs on CPU (`linear\_evals\_043.py`, results in `results/linear\_evals\_043.json`; neither exists yet). The reason is the 031 operator audit: on single deletions the transformer's score was a learned affine read of the deleted gene's own token, reproduced by a ridge from that row, and the ridge on the raw ProtT5 token read 0.082 on the same pin. The ridge from the deleted gene's frozen composite token to its expression profile is therefore the real split-A bar, and it has never been measured on expression. 035 found nested ridge on the raw fingerprint, 0.303, the bar no model beat with an interval clear of zero; 030 found a zero-parameter lower-order null at 0.577 above the joint transformer's 0.562.

Strain vectors. A single deletion is the deleted gene's frozen composite token x_g (Sec. 7.2), and each member of it alone, so the ladder says which block carries what. A double is $x_a + x_b$ and $[\,x_a + x_b\,;\,x_a \odot x_b\,]$; since no double trains, the inductive additive null is $f(x_a) + f(x_b)$ from the single-fit ridge. An isolate is, in order of richness, the 0/1 vector of its absent and loss-of-function genes, the genotype kinship matrix over the 979 isolates from the sequence-variant calls (kinship ridge is the GBLUP bar), and the change-type-weighted sum of x_g over its marked genes. A segregant is its parent-of-origin vector over genes (+1, −1, 0 for unassigned) from the posterior 0.9 blocks. A promoter is its one-hot 80-mer and k-mer counts, $k \le 6$, and the frozen f_{up} window once the plasmid is in.

Rungs. N0, the per-gene mean, the training-mean profile for every strain; ridge, multi-output and closed form, penalty on the inner fold, inputs and targets z-scored on training rows; kNN, the mean profile of the k nearest training strains by cosine on x_g, $k \in \{5, 10, 20\}$; GBLUP for isolates; the marker ridge for segregants; the k-mer ridge for promoters; and for the doubles a four-rung ladder that sees only the double's components: the per-gene mean, the stronger single, $M_x + M_y$, and the fitted $aM_x + bM_y + c$ (leave-one-out over the 72). Every difference is paired by held-out unit with a 95% bootstrap interval resampled over strains (per-strain statistics) or genes (per-gene statistics); the strain interval

is the one quoted, as the compound interval was in 035. A ridge over 1,400 strains by 3,328 inputs to 6,169 targets is one solve per penalty per fold, seconds on CPU; the whole ladder is a CPU job.

What would say the embedding carries the signal. Per dataset, ridge or kNN on x_g above N0 on the median per-gene Pearson with the strain interval clear of zero. For Kemmeren the reference points are the 019 window mean 0.180 ± 0.006 and the ceiling 0.729; no number is predicted for the token ridge. For the isolates, GBLUP against N0 on Muenzner is the bar the transformer must clear, with Caudal against Muenzner (0.208 per isolate, 0.160 per gene, Table 7) the reference for another readout of the same strain. For the doubles, a model counts only above the fitted combination with at least 5 of 6 seeds up, on the 72 and on the restricted genes.

8.6 The evaluation protocol ✗

One fold file, five held-out sets, one probe, the arms and a scoring rule, declared before any run is read (Table 19). The 019 window rule stands: budget and window declared with the launch (500 epochs read over epochs 400 to 499 is the precedent; its 1,000-epoch runs scored lower over 900 to 999 than over 400 to 499 in 7 of 8), a cell read by its escape fraction first and its window mean second, no maximum over epochs. Two rules are added from 031 and 035: the validation fold monitors the escape and never selects the epoch (031 found epoch selection on validation null, -0.009), and the test fold is read once per round. Spearman is reported beside Pearson everywhere, and every calibration slope is named by its population (the whole fold and the predicted top decile), since 010 measured 0.88 on a whole test split against 1.59 inside the predicted top 100.

The fold file. `conf/splits/global\_split\_v1.json`, written by `make\_global\_split\_043.py` with `default_rng(0)` from the raw store's index and validated by the pydantic record `GlobalSplit043` (version, seed, store sha256, test fold 0, validation fold 1, the pinned training genes, and one fold id per key in four key spaces). Deletion genes: every gene deleted in Kemmeren, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak or Ohya, the 82 Sameith singles pinned to the training folds (P2), the rest in ten folds by a draw stratified on the gene's dataset-membership pattern, so each dataset holds out about a tenth of its genes (Table 11). A deletion record trains only if none of its genes is in fold 0 or 1. Isolates: the 979 strain names of Caudal and Muenzner (759 shared, 55 with turnover) in five folds stratified by membership and turnover. Segregants: five folds within each cross, de novo cells as records of their own. Promoters: five folds over distinct Vaishnav 80-mers. Ten folds for genes and five for the rest because a held-out gene leaves every deletion dataset at once and Kemmeren at 1,400 eligible genes cannot spare 40% to test and validation. The dataset module answers `role(sample)` from the file; a per-dataset-alone run is the joint configuration with one dataset listed, so alone and joint read the same held-out records. Every run logs the file's version and store hash and refuses a mismatch. A second, clade-disjoint isolate file is the hard split once the Peter 2018 clade is on the isolate descriptor (owner question 6); it sits beside the random file and does not replace it.

Split A, deletion genes. Fold 0 of the gene folds, scored on every deletion dataset that measures the gene. The primary statistic is the median per-gene Pearson across the held-out strains; the per-strain Pearson across genes is reported centered on the training-mean profile, with the raw value and the Kemmeren different-mutant floor of 0.284 beside it (job 5893; Jackson's different-genotype constructions agree at 0.095). The bars are the ladder of Sec. 8.5 on the same genes; the ceiling is Kemmeren's own-culture 0.729 (test-retest) and 0.653 (decomposition); the 019 by-strain reading 0.180 ± 0.006 is a reference and not a paired bar. A held-out gene still appears in the isolates and segregants as a sequence variant, an absence or an inherited block; those are not deletions and are not held out. Jackson is scored only inside the joint arm, never alone.

Split B, every Sameith double. All 72 Sameith 2015 double deletions are held out; their 82 constituent singles are in training as Kemmeren profiles (Sameith Sm is excluded outright, the same arrays). Each double is scored by the centered per-strain Pearson against its measured profile and paired against the four rungs of Sec. 8.5, the additive null being the paper's own expected value ("the deviation between the expression change observed in the double mutant $M_{x\Delta y\Delta i}$ and the expected expression change, given each single mutant $M_{x\Delta i} + M_{y\Delta i}$", paper.md line 39); the same on the genes where $|M_{xy,i} - (M_{x,i} + M_{y,i})| > \log_2 1.5$, where the additive null is wrong by construction. Sameith ran "four times from two independent cultures" per mutant (paper.md line 33) and released one profile, so no double has a replicate; the Kemmeren 0.729 is the bound for the same arrays.

Split C, isolates. Fold 0 of the isolate folds, out of Caudal, Muenzner protein and turnover together. The primary statistic is the median per-protein Pearson across the held-out isolates over 1,540 proteins; the per-isolate Pearson is reported centered. GBLUP on the same isolates is the bar; Caudal against Muenzner is the reference for another readout of the same strain; the turnover read is the fold average over all five folds, the only stable read at 55 isolates. The random isolate fold is the easy split, since relatives of a held-out strain sit in training, and GBLUP will look strong on it.

Splits D and E, segregants and promoters. Fold 0 within each cross, scored by the median per-gene Pearson

across held-out segregants against the marker ridge; fold 0 of the promoters, scored by Pearson and Spearman across held-out promoters against the k-mer ridge.

The transfer statistic and the rule. For dataset d and initialization seed s, $T_d(s) = S_d(\text{joint}, s) - S_d(\text{alone}, s)$ on the fold-0 records of d, with S_d the primary statistic. Alone and joint are matched on optimizer steps, not epochs, since an alone run sees fewer records per epoch. A gain, transfer or arm difference is called when its mean is at least 0.02 on the primary statistic, at least 5 of 6 paired seeds are up, and the 95% interval resampled over held-out strains (genes for per-gene statistics) excludes zero: 0.02 is three seed standard deviations on either readout (019 expression 0.006 over four runs, proteome 0.008 over twelve), 5 of 6 is the 031 campaign rule at six seeds (9 of 12 at twelve), and the interval guards against six seeds agreeing on the sign of a 0.005 effect. The rule is committed as `conf/decision\_rule.yaml` before the first GPU run. Six seeds are six initializations on the pinned fold 0; fold 1 as test with fold 0 as validation is the second split seed when the budget allows, and the rule then reads 9 of 12. One seed is never read.

The morphology probe. Ohya 2005 enters no training. For each of its 4,678 genotypes the frozen model produces h, z_S and the mean over genes $\bar{H}$; a ridge from each, from their concatenation and from the frozen token x_g to the 116 moving CalMorph features is fit 5-fold over genotypes with the penalty chosen on an inner split, inputs and targets z-scored and clipped at 8, and scored as the out-of-fold Pearson per feature across held-out genotypes, median over the 116 (the construction of Table 8). Two reads: raw, and growth-residualized, with the Costanzo 2016 single-mutant fitness of the genotype regressed out of every feature on the training folds and appended as a covariate, because the chemogenomic growth matrices predict morphology nearly as well as the proteome (Hoepfner 0.263, Hillenmeyer 0.246, Messner 0.282, three growth panels 0.310; 019 screen, job 3919). The bars, in order: the 019 genotype-only model read through its own trained head, 0.149 ± 0.022; the sequence-embedding ridge 0.08 to 0.14 (`11-inductive.tex`); the Kemmeren-expression ridge 0.348 on the 1,437 shared genotypes, the Messner ridge 0.278 on 4,279 and the Hu ridge 0.235 on 259 (Table 8, job 5892), transductive and each also paired on its own strains; the residualized bars recomputed in the same script; and the ceiling 0.847 on the 116 or 0.611 over the 281. The learned state carries morphology beyond growth when the residualized median clears the residualized Kemmeren ridge with the strain interval clear of it; a raw median near 0.149 says the probe is the trained head by another route. The Ohya genotypes whose gene is in fold 0 (about 468) are the strictly inductive subgroup, unseen on every readout. Hypothesis (untested): z_S carries the most, since the 031 audit placed the score in the deleted gene's own row and z_S is the readout 019 trained.

Arms. Main: every training dataset, heads per level, tokens at the head with the per-dataset bias, HL-Gauss. Data arms in a 2×2: with and without the natural-genome datasets (Caudal, Muenzner protein and turnover, Boocock, N'Guessan), with and without Vaishnav (once its window exists); the arm without Hu, Jackson and Nadal-Ribelles, which can only lower the Kemmeren score if their signal does not transfer (owner question 5 names the headline control). Loss arms: pinball; one MSE cell. Token arms: the attended layer (arm 1); the flat type id; no token. The six-heads-by-type control; the record-count weighting arm; the transductive identity-table ceiling; and the alone runs, one per training dataset. Every arm is scored on splits A and B and the probe, C, D and E in the arms that train those datasets; six initializations per cell on fold 0. Main plus these arms at six seeds is on the order of 70 runs of 500 epochs before the alone runs, so the linear ladder cuts any arm whose lower-order bar already answers it, and the owner ranks the rest.

8.7 Owner decisions ✗

Each item names the main option first; Table 12 carries the same rows with their measured facts.

1. **Heads.** One head per readout level, shared across readout types and conditioned by the type token, on targets standardized per dataset with a per-dataset bias (c_d on the logits, b_d under pinball or MSE). Control arm: six heads by type.
2. **Experiment-type token.** Ontology path with the modeling-side grouping (expression, protein, promoter, fitness, interaction, metabolite, morphology); the flat id as the ablation.
3. **Where the tokens enter.** At the head (main); type and environment attended in one layer after the operator (arm 1); no token (ablation). The dataset never enters the trunk.
4. **Loss.** HL-Gauss bins ($K = 32$ over $[-4, 4]$ in z units, σ 0.75 bin widths, expectation as the point); the 19-quantile pinball as arm 1; one MSE cell as the reference.
5. **Dataset weighting.** Equal per dataset; agreement weighting dropped; record count as an arm.
6. **Nadal-Ribelles sentinels.** $|\log_2 \text{FC}| > 20$ set to missing before standardization (20.2% of cells).
7. **Jackson target.** The effect against the HO control of the same environment; Jackson not scored alone.

Table 19: The evaluation protocol. Every bar is a measured number or a rung computed on the same held-out units; a reference is a number from a different split or model and is marked so. Held-out sizes are expected fold-0 shares (Table 11); per-strain Pearson is reported centered on the training-mean profile, the raw value beside it.

Split	Held out	n	Primary statistic (secondary)	Bar
A, deletion genes	gene fold 0 of the fold file, out of every deletion dataset; the 82 Sameith singles pinned to training	about 140 Kemmeren strains, 448 Messner, 306 Nadal, 26 Hu, 10 Zelezniak; 6,169 genes	median per-gene Pearson across held-out strains (centered per-strain Pearson, median; raw beside it)	ladder on the same genes: per-gene mean, ridge and kNN on the frozen token; ceiling 0.729 test-retest, 0.653 decomposition; raw per-strain floor 0.284; reference 019 by-strain 0.180 ± 0.006
B, Sameith doubles	all 72 double deletions; the 82 singles in training as Kemmeren records	72 strains, 6,169 genes	centered per-strain Pearson vs measured; paired difference vs four rungs; the same on genes with $\lvert M_{xy} - (M_x + M_y)\rvert > \log_2 1.5$ (Spearman; slope in the predicted top decile)	per-gene mean; stronger single; $M_x + M_y$; fitted $aM_x + bM_y + c$ (leave-one-out); ceiling 0.729
C, isolates	isolate fold 0, the same strains out of Caudal, Muenzner protein and turnover	about 196 isolates (about 159 Muenzner, 11 turnover), 1,540 proteins	median per-protein Pearson across held-out isolates (centered per-isolate Pearson; turnover as the fold average)	GBLUP on the same isolates; reference Caudal × Muenzner 0.208 per isolate, 0.160 per gene; reference 019 Messner 0.085 ± 0.008
D, segregants	fold 0 within each cross	about 15,903 Boocock records, 590 N'Guessan	median per-gene Pearson across held-out segregants (centered per-segregant)	marker ridge on the parent-of-origin vector; nearest-genotype r 0.113 vs 0.070 random (Sec. 3.1)
E, promoters	fold 0 over distinct promoter variants	about 14,232 promoters	Pearson across held-out promoters (Spearman)	k-mer ridge; one-hot ridge; frozen-window ridge once the plasmid is in
Morphology probe	Ohya 2005, never trained; 5-fold over genotypes; fold-0-gene subgroup	4,678 genotypes, 116 moving features	ridge from frozen h, z_S, $\bar{H}$, their concatenation, x_g; out-of-fold Pearson per feature, median over 116; raw and growth-residualized	019 trained head 0.149 ± 0.022; sequence ridge 0.08 to 0.14; Kemmeren ridge 0.348 (1,437), Messner 0.278 (4,279), Hu 0.235 (259), transductive, residualized twins recomputed; ceiling 0.847 (116) or 0.611 (281)

8. **Split A leakage rule.** The held-out gene out of every deletion dataset, from the fold file.
9. **Split C.** Fold 0 of the isolate folds, about 196 isolates with about 159 Muenzner and 11 turnover; random over isolates now, a clade-disjoint second file when the clade is on the descriptor.
10. **Budget, window and selection.** Declared with the launch; alone and joint matched on optimizer steps; the validation fold never selects the epoch; the test fold read once.
11. **The fold file, the rule, the ladder, the centered statistic, the transfer statistic, the probe control and the reporting** are rows S1 to S7 of Table 12.